

**AFFECTIVE DECISION-MAKING IN RHESUS MONKEYS:
REWARD ORDER, RISK, AND THE ROLES OF THE
ANTERIOR INSULA AND PREMOTOR CORTEX**

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ABSTRACT

From deciding what to eat first or what stock to invest in, the order in which we prefer to experience events and the amount of risk we are willing to undertake are factors that influence our choices. However, the mechanisms underlying order and risk preferences remain unclear. This thesis presents experiments with rhesus monkeys (*Macaca mulatta*) designed to delineate these mechanisms and their evolutionary trajectory. The behavioral findings show that rhesus monkeys usually prefer receiving a high quality food-item first in a sequence of rewards, similar to many people. Moreover, similar to people, the monkeys are able to integrate value and likelihood to guide their choices.

Regarding the brain mechanisms underlying risky decision-making, previous studies have shown that brain regions like the anterior insula and premotor cortex are important for affective decision-making. Damage to the anterior insula cortex, a brain region believed to integrate homeostatic urges with cognition, decreases addictive urges. The premotor cortex represents motor task set and is one of the output structures for choice behavior. However, exactly how decision-making variables such as expected value and risk are encoded by individual neurons in these brain regions remains unclear. To address this, I recorded neurons from both the anterior insula and the premotor cortex while two rhesus monkeys performed a gambling task in which the monkeys chose between safe and risky options for fluid reward. Unlike anterior insula neurons, the population of premotor neurons was significantly more location selective before and after choice. Neurons from both brain regions, however, were tuned to decision-making variables such as expected value, risk, and relative value. Compared to premotor cortex, anterior insula neurons showed an increased coding of risk during choice and of positive

expected value after the choice was made during reward anticipation.

Altogether, my experiments show how the order of reward-items, the risk, and the expected value associated with choice options influence affective decision-making in rhesus monkeys. Furthermore, the premotor and anterior insula cortices both appear to play important and separable roles in making and evaluating affective decisions.

PREFACE

Professionally, I must thank the Psychological and Brain Sciences (PBS) department here at Dartmouth College for allowing me to be a graduate fellow in this wonderful program with outstanding resources, facilities, and staff. I would like to thank Professor Duchaine, the PBS graduate program chair, for earnestly and patiently keeping an eye on my progress in my final year at Dartmouth. I also thank all present and past members in my lab for all types of experimental assistance not limited to invaluable scholarly input. This work would not be possible without the staff and vets from the Center for Comparative Medicine & Research (CCMR) at Dartmouth who helped with all aspects of ensuring our monkeys' health. My committee members have also been wonderful: Dave Bucci, Paul Whalen, Antoine Bechara, and my P.I. and chair of the committee, Jerald Kralik. I am truly fortunate to have all of you.

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This thesis is dedicated to our laboratory monkeys: Caesar, Hamlet, Puck, and Titus.

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Chapter 1.

Affective Decision-Making in Rhesus Monkeys

1A. The Effect of Reward Order

1A.1. INTRODUCTION

According to utility maximization theory, individuals should prefer options that maximize overall utility regardless of the order of outcome events (D. Kahneman & Tversky, 1979). However, Kahneman and colleagues (1993) showed that people preferred a longer duration of a painful water bath that ends slightly less painfully to a shorter one that does not change in pain intensity, even when the former option entailed more cumulative pain overall. Similarly, Redelmeier and colleagues (Redelmeier & Kahneman, 1996; Redelmeier, Katz, & Kahneman, 2003) showed that colonoscopies were rated less unpleasant when they lasted longer but ended more gently to those that were significantly shorter but did not change in intensity. Aversive loud noises were also rated as less negative when ending on a less aversive note (Schreiber & Kahneman, 2000). These findings run counter to the principle of maximizing overall utility and suggest a peak-end rule in humans whereby alternatives that end less negatively are preferred, even at the expense of overall utility maximization.

The examples mentioned so far involve negative, aversive stimuli, but positive stimuli show similar effects. Both adults and children (elementary and middle school) subjectively rated their experience higher when receiving a high-valued reward by itself (a highly-rated DVD movie for adults, candy for children) against a higher valued reward followed by a lower valued reward (a lower quality movie for adults and a less desired candy for children; (Do, Rupert, & Welford, 2008)). In another study, people rated advertisements more positively when the interval length between their highest positive feelings for the advertisement and the end of the advertisement was shorter, and

they rated them more negatively when the interval length was longer (Baumgartner, Sujan, & Padgett, 1997). These results again suggest a peak-end preference whereby people rated a subjective experience based on how well the experience ended, even at the expense of overall utility. Although significant differences have been found between testing paradigms that use choice preferences versus subsequent outcome evaluations (see (Wilson & Gilbert, 2005)), the peak-end effect appears to hold for both ratings (Baumgartner, et al., 1997; Do, et al., 2008; Redelmeier, et al., 2003; Schreiber & Kahneman, 2000) and choices (Jeong, Oh, Chun, & Kralik, 2014; D. Kahneman, et al., 1993; Redelmeier, et al., 2003).

To determine the generality of order preferences among primates, it is important to conduct a comparative analysis with other species. Some chimpanzees have been found to exhibit a *selective value effect* whereby individuals appeared to value equally a higher valued food item in isolation and the same higher valued item in conjunction with a lower valued one. The suggestion was that the lower valued food item was of no value in the presence of the better one (Silberberg, Widholm, Bresler, Fujita, & Anderson, 1998). However, it was unclear whether there was any order preference since the food items were offered simultaneously. Furthermore, chimpanzees switched their preferences to the food mixture when the study was conducted with equalized trial lengths and longer delays between trials (Beran, Ratliff, & Evans, 2009), suggesting utility maximization trial by trial when the chimpanzees were unable to receive the preferred item in quick succession across trials. Notably, however, when the chimpanzees chose the food mixture, they tended to eat the preferred food item first the majority of the time (Beran, et al., 2009).

Heightened impulsiveness and sensitivity to the higher valued rewards might imply a peak-first preference when goods are obtained sequentially due to the preferred food dropping in value the longer a subject has to wait for it (Hayden & Platt, 2007). However, it is possible that there is something special about the order of receiving outcomes that is independent of delayed discounting and risk assessment. People use aphorisms such as “save the best for last” or “all’s well that ends well”, which reflects the effect of order independent of time or the risk of losing the final reward. Blanchard and colleagues (Blanchard, Wolfe, Vlaev, Winston, & Hayden, 2014) recently demonstrated an effect of order by finding that rhesus monkeys can show a peak-end preference under their task parameters, challenging reward discounting models. The authors trained monkeys to choose to whether to repeat or switch from a given reward sequence consisting of five squirts of juice reward. The five squirts always contained four squirts of equal volume of juice and one squirt with a higher volume of juice. By varying when in the sequence the large volume squirt was given, the authors found that the monkeys stayed most often with a sequenced reward that ended with the higher fluid reward, i.e. a peak-end order preference. Although the study showed that monkeys behave similarly to humans who show a peak-end preference, the short delay between choices and the manipulation of reward value with quantity (as well as other explanations) may have contributed to the results (see **1A.8.GENERAL DISCUSSION**).

Overall, then, there is evidence for order preferences in humans and some nonhuman primates, but the effects are mixed and may depend on task-related parameters. In the current study, we tested rhesus monkeys to determine whether they would exhibit an order preference between qualitatively different food items with a

significant intertrial interval, equalized trial lengths, and a lower number of trials per session. Due to heightened impulsiveness, sensitivity to large rewards, and the salient differences between the food items in the sequence, we hypothesized that the rhesus monkey choices would be affected by order in our experiments, and, furthermore, that they would more likely exhibit a peak-first preference— that is, preferring to receive the best item first rather than last.

1A.2. GENERAL METHOD

If A represents the higher valued item, and B represents the lower valued one, we represented the sequence of receiving A then B as $A \rightarrow B$, and the sequence of receiving B then A as $B \rightarrow A$. We tested $A \rightarrow B$ versus $B \rightarrow A$ directly to determine whether order of outcomes influenced rhesus monkey choice behaviour. We hypothesized that all monkeys would prefer $A \rightarrow B$ over $B \rightarrow A$.

1A.2.1 Subjects

We tested three male rhesus monkeys, denoted as monkey T, P, and H, ages 9, 7, and 7 years, respectively. All monkeys were maintained at approximately 95% of their ad libitum weights. The monkeys were housed in a homeroom and were rolled to the testing room in our laboratory in custom-made chairs. All three subjects were used in one response time and one forced-choice behavioural experiment (Knight, Klepac, & Kralik, 2011). All experiments in this report complied with all current laws and regulations of the United States, the United States Department of Agriculture (USDA), and the Institutional Animal Care and Use Committee (IACUC) of Dartmouth College for the use of animals in research.

1A.2.2 Materials

In the test room the monkey's chair was placed across a table (30 inches) from the experimenter. The monkey sat in the chair, with his left arm loosely restrained (using a custom-made metal ring attached to the chair around both the upper and lower arms) and the right arm free to reach. The custom-made chairs were used for tighter experimental control over behaviour. Two grey opaque canisters were used as stimuli. Using construction paper covered by clear tape, we changed each canister's lid to a particular color that was easily viewable by the monkey. The color corresponded to each reward contingency described below (see specific methods and tables for all experiments). In Experiments 1–4 for monkeys P and H, and Experiments 1 and 4 for monkey T, a button panel consisting of two unlit buttons was centered in front of the monkey. We removed the button panel for monkey T in experiments where he exhibited prolonged side biases during learning of the reward contingencies (removing the panel relieved these biases). We removed the button panel for all monkeys in Experiment 5. The monkeys made selections by touching one of the two buttons corresponding to the canister lid presented across from it. When the button panel was removed, the monkeys made selections by directly touching the canister's colored lid. All experimental sessions were videotaped.

1A.2.3 Procedure

We tested each monkey on one session per day. Each session consisted of 10 familiarization trials and 30 experimental trials. With this total number of trials, there was no evidence of satiation effects (all monkeys ate the food items in all sessions). The

purpose of familiarization was to allow the monkeys to learn the colored lid and reward contingencies, which did not change within an experiment. During familiarization, one canister was presented per trial, randomized among the two different colors (for color and reward contingencies see tables within each individual experiment) in the experiment and two different positions (left or right). This procedure was modified for monkey T in Experiments 2, 3, and 5 to facilitate the learning of the reward contingencies: Two canisters were used in familiarization, with the second one having a red lid and yielding no reward upon choice. Upon choice in both procedures, the canister's lid was pointed back towards the experimenter so the food within it was not viewable to the monkey. The lid was then removed, and the monkey received the contents based on the experimental contingencies. The intertrial interval was approximately 8 s.

All experimental trials were conducted using a two-alternative forced-choice procedure. Each trial began when the experimenter held two closed opaque canisters 9 inches from the edge of the button panel, or 18 inches from the monkey if the button panel was not used. We held the canisters at an angle so that the colored lids were easily viewable to the monkeys. We held them in place until the monkeys had looked at the alternatives for approximately 3 s, then brought them forward to the edge of the button panel or within reaching distance of the monkey if the button panel was not used. Canister positions were pseudorandomized so that the same positions did not occur more than twice in a row.

To minimize potential inadvertent cues from the experimenter, several procedures were implemented: (a) White noise was played to minimize any auditory distractions; (b) the experimenter wore a white lab coat, mask, safety goggles, and medical gloves

throughout the session; (c) five different experimenters tested the monkeys, following a stereotyped procedure, and we obtained no differences between experimenters.

Three significant sessions (two-tailed binomial tests per session, $p < 0.05$) or 10 consecutive nonsignificant sessions were required from each monkey before moving to the next experiment. The three significant sessions were not necessarily consecutive. We therefore considered the overall preference to be the average over all trials from the first significant session to the last significant session. We started with the first significant session because in some cases the monkeys needed extra time to learn the reward contingencies associated with each color. Once there was one significant session, we obtained two more to verify stability. Ten consecutive nonsignificant sessions were taken to mean there was no preference by the monkey.

Food preferences were conducted to determine the superior/inferior food items. It was important for the inferior item to be consumed when presented alone so that it was not aversive and was of positive value. Because monkey P did not eat sugar snap peas as readily as monkeys T and H, green beans were used for that subject. After informal preference testing to determine the best food items to use (selecting a highly valued and lower valued item) prior to the study, we conducted one formal 30-trial session per monkey to verify preferences. All monkeys selected one grape over a half vegetable (green bean for monkey P, sugar snap pea for monkeys H and T) on every trial. We denoted the superior item (the grape) with the variable “A” and the inferior item (a half vegetable) “B”.

1A.3. EXPERIMENT 1

1A.3.1 Method

Subjects, materials, and procedure

These were as listed under “General Method”.

1A.3.2 Specific procedure

Selection of one colored lid resulted in the monkey receiving a grape, followed 4 s later by a half vegetable, denoted as $A \rightarrow B$. The 4-s delay was added to provide time for the monkey to consume the food item. The other colored lid gave a half vegetable, followed 4 s later by a grape, denoted as $B \rightarrow A$. Table 1 shows lid colors and contingencies for each monkey.

Monkey	Canister colors/contingency	
	$B \rightarrow A$	$A \rightarrow B$
P	Purple	Yellow
H	Green	Blue
T	Purple	Blue

Table 1. Reward order experiment (1A): The Canister Lid Colors and Contingencies for Experiment 1.

Note: A monkey selected from two canisters with colored lids. Choice of one canister resulted in a half vegetable followed 4 s by a grape ($B \rightarrow A$), and choice of the other resulted in a grape followed 4 s later by a half vegetable ($A \rightarrow B$). Monkeys were given different colors in the experiments to avoid color confounds.

1A.3.3 Results

As shown in Figure 1, all three monkeys (P, H, and T) exhibited a significant preference for $A \rightarrow B$ over $B \rightarrow A$, consistent with a peak-first rather than peak-end preference. All three sessions for monkey P were significant, selecting $A \rightarrow B$ on 70%, 86.7%, and 86.7% of the trials (two-tailed binomial tests: $p < 0.05$, $p < 0.0001$, $p < 0.0001$). Overall, monkey P selected $A \rightarrow B$ on $81.1 \pm 9.6\%$ of trials across the three sessions (X^2 : $p < 0.001$). All three sessions for monkey H were also significant, selecting $A \rightarrow B$ on 70%, 73.3%, and 70% of the trials (two-tailed binomial tests: $p < 0.05$, $p < 0.05$, $p < 0.05$). Overall, monkey H selected $A \rightarrow B$ on $71.1 \pm 1.9\%$ of trials across the three sessions (X^2 : $p < 0.02$). Monkey T took four sessions to have three significant ones, selecting $A \rightarrow B$ on 70%, 76.7%, and 70% of the trials (two-tailed binomial tests: $p < 0.05$, $p < 0.01$, $p < 0.05$). Overall, monkey T selected $A \rightarrow B$ on $70 \pm 5.4\%$ of trials across the four sessions (X^2 : $p < 0.02$).

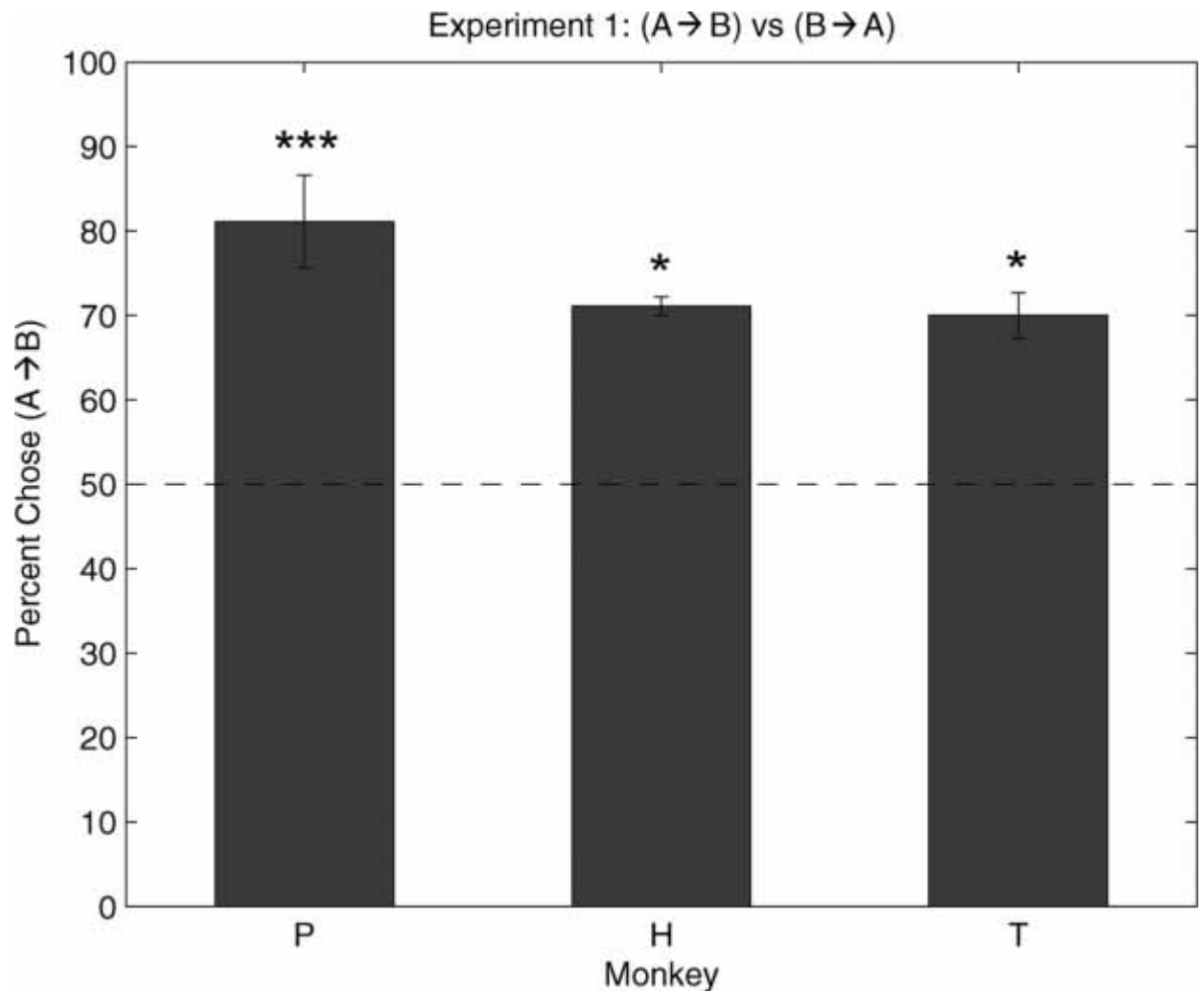


Figure 1. Reward order experiment (1A): Experiment 1 results, with one choice giving a grape followed 4 s later by a half vegetable ($A \rightarrow B$), and the other giving a half vegetable followed 4 s later by a grape ($B \rightarrow A$). The y axis plots the mean percentage of trials that each monkey selected ($A \rightarrow B$) from the first significant session to the third significant session. Error bars are standard errors of measurement, the dashed line at 50% is random performance, and asterisks represent each monkey's significant chi-squared (X^2) p-value across the sessions: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

1A.3.4 Discussion

Rather than displaying a possible peak-end or recency effect by preferring the outcome with the last and most recent best experience, the monkeys clearly preferred the alternative with the best item received first. This result potentially contrasts with humans, in which we appear to have a peak-end, recency preference. Given that rhesus monkeys may display the selective value effect and ignore the lower valued item (B; (Silberberg, et al., 1998)), it is not clear how much the choice behavior of any of the monkeys can be explained by order versus simply the delay of the preferred food item (A). In Experiments 2 and 3, we attempted to determine whether the lower valued item (B) affects choice in both the second and the first positions.

1A.4. EXPERIMENT 2

In Experiment 1 we tested two different outcome orders against each other directly: $A \rightarrow B$ versus $B \rightarrow A$. However, it was important to determine whether both food items in the sequence actually affected choice. Thus, in Experiments 2 and 3, we attempted to determine whether the lower valued item affects choice both in the second and in the first positions. In Experiment 2, we determined the value of B in the second position by comparing $A \rightarrow B$ to A alone. If B is of positive value, $A \rightarrow B$ should be preferred to A alone. If B is of zero value and not affecting choice, removing it should have no effect. That is, the monkeys should be indifferent to the choice alternatives. If B has negative value, A alone should be preferred. This last finding would be comparable to that of Do et al.(2008), in which people preferred A alone to $A \rightarrow B$.

1A.4.1 Method

Subjects, materials, and procedure

These were as listed under “General Method”.

1A.4.2 Specific procedure

Choice of one colored lid resulted in the monkey receiving one grape, followed 4 s later by a half vegetable (sugar snap pea or green bean), denoted $A \rightarrow B$. Again, the 4 s allowed the monkey to eat the grape before receiving the vegetable. The other colored lid led to the monkey receiving one grape, followed by a 4-s delay, denoted as $A \rightarrow []$. We added the delay to maintain comparable trial lengths for both choice alternatives. Table 2 shows the lid colors and contingencies for each monkey.

Monkey	Canister colors/contingency	
	$A \rightarrow []$	$A \rightarrow B$
P	Blue	Yellow
H	Yellow	Blue
T	Yellow	Blue

Table 2. Reward order experiment (1A): The canister lid colors and contingencies for Experiment 2

Note: Choice of one canister resulted in a grape followed by a 4-s delay ($A \rightarrow []$), and choice of the other resulted in a grape followed 4 s later by half a vegetable ($A \rightarrow B$).

1A.4.3 Results

Two out of three monkeys (P and H) showed a significant preference for $A \rightarrow B$ over $A \rightarrow []$ (see **Figure 2**). All of monkey P's sessions in this experiment were significant, selecting $A \rightarrow B$ on 90%, 83.3%, and 73.3% of the trials (two-tailed binomial tests: $p < 0.00001$, $p < 0.001$, $p < 0.05$, respectively). Overall, monkey P selected $A \rightarrow B$ on $82.2 \pm 8.4\%$ of all trials across the 3 sessions (X^2 : $p < 0.001$). All of monkey H's sessions were also significant, selecting $A \rightarrow B$ on 76.7%, 93.3%, and 86.7% of the trials (two-tailed binomial tests: $p < 0.01$, $p < 0.00001$, $p < 0.0001$). Overall, monkey H selected $A \rightarrow B$ on $85.6 \pm 8.4\%$ of trials across the 3 sessions (X^2 : $p < 0.001$). Because monkey did not have even 1 significant session in 10 sessions, we concluded that he did not have a preference. Monkey T selected $A \rightarrow B$ on average $51.1 \pm 10.2\%$ of trials over his last 3 nonsignificant sessions (X^2 : $p > 0.5$).

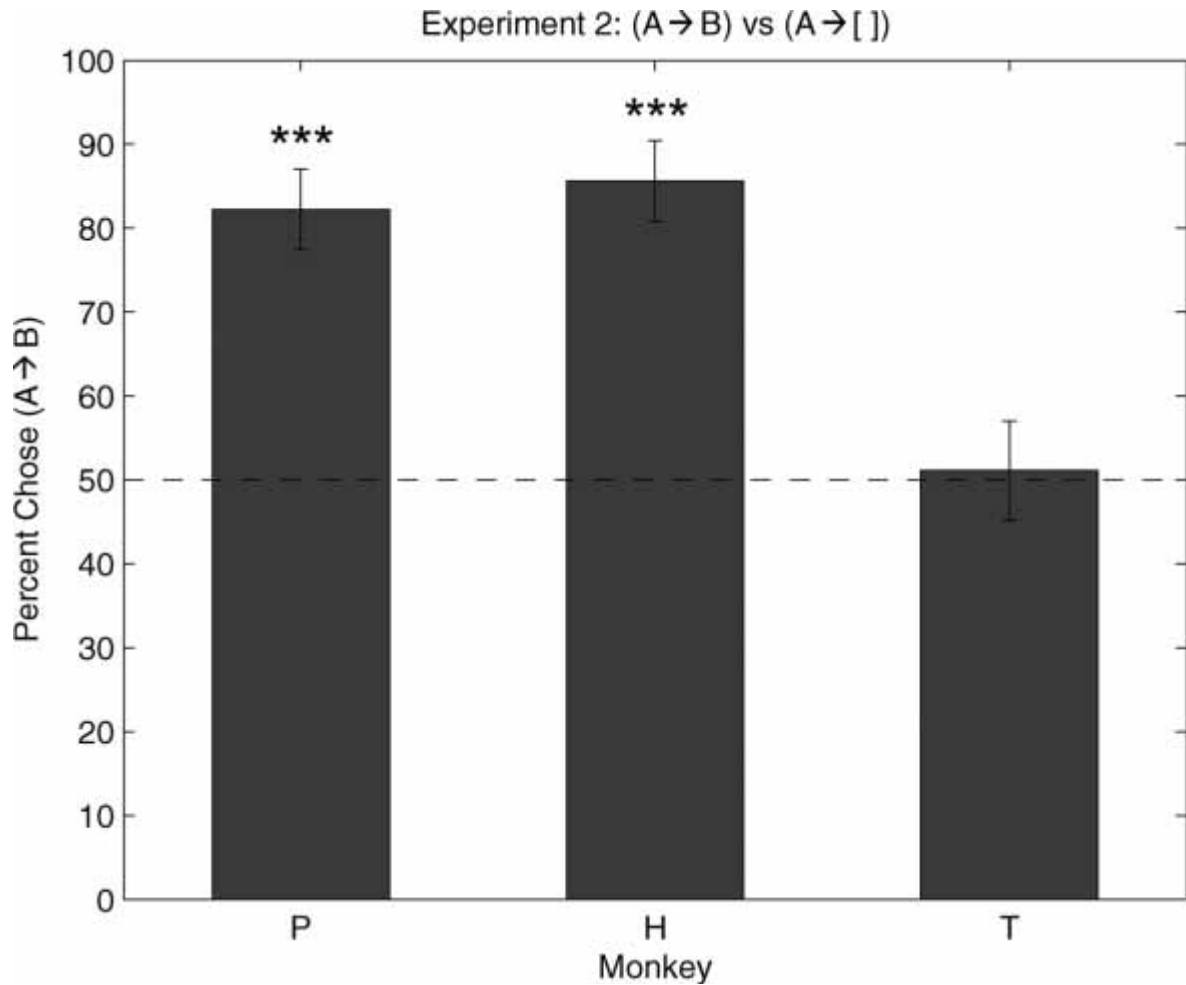


Figure 2. Reward order experiment (1A): Experiment 2 results, in which the monkeys selected from two colored canisters, with one giving a grape followed 4 s later by half a vegetable ($A \rightarrow B$), and the other giving a grape followed by a 4-s delay ($A \rightarrow []$). The y axis plots the mean percentage of trials that each monkey selected ($A \rightarrow B$) from the first significant session to the third significant session. Error bars are standard errors of measurement, the dashed line at 50% is random performance, and asterisks represent each monkey's significant chi-squared (χ^2) p-value across the sessions: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Monkey T's result is the average and standard error of measurement over the final three nonsignificant sessions.

1A.4.4 Discussion

For two monkeys, P and H, the preference for $A \rightarrow B$ over $A \rightarrow []$ showed that B influenced the choice behavior positively (or at least that the value of B was greater than the value of the short delay). And since A was more highly valued than B, both A and B ($A \rightarrow B$) influenced the behavior of these two monkeys positively, whereas, monkey T's behavior appeared to be influenced only by A, with B being essentially ignored and thus having zero value. This result appears to provide more evidence for the selective value effect, in which a lower valued food item is ignored and holds zero value when in the presence of a higher valued item (Beran, et al., 2009; Silberberg, et al., 1998), although in the current case, the effect occurred with the items presented sequentially rather than simultaneously.

Our results contrast with the peak-end findings in people in which participants significantly rated the experience higher when it was the highly valued reward in isolation (e.g., (Do, et al., 2008)). In the current experiment, none of the monkeys showed a preference for A alone. This difference could be due to a true species difference in order preferences. However, it is possible that differences in the testing procedures from that of the Do and colleagues' experiment could underlie the opposing results. In Do and colleagues' experiment, participants did not have a choice, and it is not certain that the subjects would choose the option that they would rate higher, especially with knowledge of both contingencies and experience selecting both alternatives. Conducting a choice paradigm in humans with repeated trials should help illuminate the factors influencing decision making and subjective value. In fact, interesting work with humans has shown that different results may be obtained between rating choice alternatives in isolation

versus comparing them directly (Christopher K Hsee, 1998; Christopher K Hsee, Loewenstein, Blount, & Bazerman, 1999; Christopher K Hsee & Zhang, 2004; List, 2002). For example, Hsee (1998) found that a 24-piece dinnerware set was rated more highly than one with the same 24 pieces, plus several more pieces, including ones that were broken. However, this preference reversed when the two options were directly compared. One therefore might predict that when given a choice, people will tend to select the alternative with the largest overall utility when making a direct comparison between $A \rightarrow B$ and A alone, especially when a fixed delay is included to eliminate receiving a quick succession of A s over trials (see (Beran, et al., 2009)). However, in the Kahneman et al. (1993) study, in which a peak-end effect was found when placing one's hand into painfully cold ice water, and in the Redelmeier et al. (2003) study, in which people received colonoscopies, the participants were given a choice between two alternatives, and they still chose the option with more overall pain (less overall utility) that ended better. Thus, our results may also suggest a possible difference between humans and another primate species such that rhesus monkeys do not show a clear peak-end preference, at least under our test conditions.

1A.5. EXPERIMENT 3

We next attempted to determine the value of B in the $B \rightarrow A$ alternative, whether being positive, zero, or even negative. To do this, we presented the monkeys with a choice between $B \rightarrow A$ and A alone, with A being presented at the same delay. If the value of B was positive, $B \rightarrow A$ should be preferred; if zero, the monkeys should be indifferent to the two alternatives; if negative, A alone should be preferred.

1A.5.1 Method

Subjects, materials, and procedure

These were as listed under “General Method”.

1A.5.2 Specific procedure

The selection of one colored lid resulted in the monkey receiving a half vegetable, followed 4 s later by a grape, denoted $B \rightarrow A$. Selection of the other colored lid resulted in a 5-s delay, followed by a grape, denoted $[] \rightarrow A$. The 5-s delay ensured that the grape was given at the same delay for either choice alternative (1 s to hand the first food item and the 4-s delay between items). Table 3 shows lid colors and contingencies for each monkey.

Monkey	Canister colors/contingency	
	$B \rightarrow A$	$[] \rightarrow A$
P	Purple	Green
H	Green	Purple
T	Black	Grey

Table 3. Reward order experiment (1A): The canister lid colors and contingencies for Experiment 3

Note: Choice of one canister resulted in a half vegetable followed 4 s by a grape ($B \rightarrow A$), and choice of the other resulted in a delay of 5 s followed by a grape ($[] \rightarrow A$).

1A.5.3 Results

Two of the three monkeys (H and T) showed significant preferences for B→A over []→A. Monkey H took 5 sessions to have 3 significant ones, selecting B→A on 80%, 76.7%, and 86.7% of the trials in the 3 significant sessions (two tailed binomial tests: $p<0.01$, $p<0.01$, $p<0.0001$). Again, taking overall preference across all sessions from the first significant session to the last, monkey H selected B→A on $74 \pm 10.4\%$ of trials across 5 sessions (X^2 : $p<0.001$; see Figure 3). Monkey T required 12 sessions to have 3 significant ones. In 3 consecutive sessions, monkey T selected B→A on 70%, 80%, and 83.3% of the trials (two tailed binomial tests: $p<0.05$, $p<0.01$, $p<0.001$). Overall, monkey T selected B→A on $77.8 \pm 6.9\%$ of trials across the 3 sessions (X^2 : $p<0.001$). Monkey P took 6 sessions to have 3 significant ones and showed the opposite effect, selecting B→A on 30%, 23.3%, and 16.7% of trials in the 3 significant sessions (two-tailed binomial tests: $p<.01$, $p<.01$, $p<.0001$). Overall, monkey P selected B→A on $26.7 \pm 8.6\%$ of all trials across the final 4 sessions (X^2 : $p<0.01$). Thus, monkey P preferred []→A over B→A.

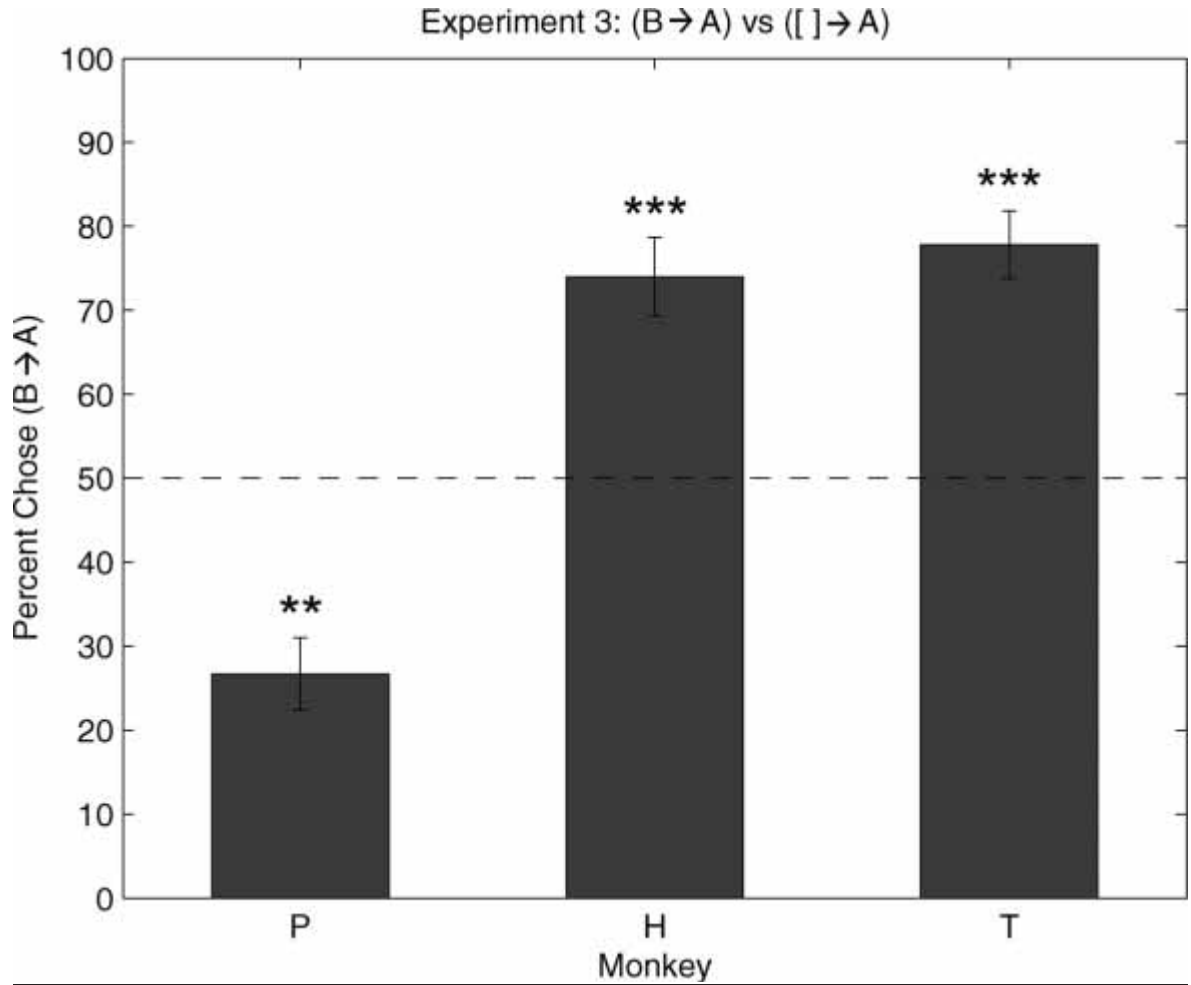


Figure 3. Reward order experiment (1A): Experiment 3 results, with one canister giving a half vegetable followed 4 s later by a grape (B→A), and the other giving an initial delay of 5 s followed by a grape ([]→A). The y axis plots the mean percentage of trials that each monkey selected (B→A) from the first significant session to the third significant session. Error bars are standard errors of measurement, the dashed line at 50% is random performance, and asterisks represent each monkey's significant chi-squared (X^2) p-value across the sessions: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

1A.5.4 Discussion

Since two of the monkeys (H and T) preferred $B \rightarrow A$ over $[] \rightarrow A$, for them the food item represented by B, the half vegetable, was of positive value when in the first position (or at least some value higher than the delay). Thus, Experiments 2 and 3 showed that for monkey H, the value of B was positive (or at least greater than the corresponding delay value in the other alternative) whether in the first or last position, and thus preferences were based on the assessment of both food items. For monkey T, however, the value of B was dependent on position, with B having no value in the last position (Experiment 1), but positive value in the first position.

1A.6. EXPERIMENT 4

To test how well the peak-first preference found in Experiment 1 generalized to other food items, we repeated Experiment 1 with two different food items. Because Experiment 1 showed that the monkeys preferred $A \rightarrow B$ over $B \rightarrow A$, we hypothesized that the same preference for receiving the best item first would hold with the new food items.

1A.6.1 Method

Subjects, materials, and procedure

These were as listed under “General Method”, except as described in the following section.

1A.6.2 Specific procedure

Grapes were replaced with mini marshmallows, hereafter called marshmallows and denoted A_2 , and the half vegetables were replaced with a single Cheerio®, denoted B_2 . For all monkeys, the marshmallows were strongly preferred over Cheerios, and the Cheerios were readily eaten when offered alone. To verify monkey preferences, we first conducted one 10-trial session per monkey. All monkeys selected one marshmallow over one Cheerio on every trial. In addition, we conducted 3 to 4 trials of preference testing on monkey P between Trials 15 and 16—that is, in the middle of the session, during each of his three significant sessions to verify that his preference did not change: He selected the marshmallow on every trial. Selection of one colored lid resulted in the monkey receiving a marshmallow immediately, followed 4 s later by a Cheerio, denoted $A_2 \rightarrow B_2$. The other colored lid gave one Cheerio immediately, followed 4 s later by a marshmallow, denoted $B_2 \rightarrow A_2$. Table 4 shows lid colors and contingencies for each monkey.

Monkey	Canister colors/contingency	
	$B_2 \rightarrow A_2$	$B_2 \rightarrow A_2$
P	Brown	Pink
H	Orange & Blue Cross	Black and white checkerboard
T	Orange & Blue Cross	Black and white checkerboard

Table 4. Reward order experiment (1A): The canister lid colors and contingencies for Experiment 4.

Note: Selection of one canister resulted in a Cheerio followed 4 s later by a mini marshmallow ($B_2 \rightarrow A_2$), and selection of the other resulted in a mini marshmallow followed 4 s later by a Cheerio ($A_2 \rightarrow B_2$).

1A.6.3 Results

Two of the three monkeys (H and T) showed a significant preference for $A_2 \rightarrow B_2$ over $B_2 \rightarrow A_2$ (see **Figure 4**). All three sessions for monkey were significant, selecting $A_2 \rightarrow B_2$ on 100%, 73.3%, and 93.3% of the trials (two-tailed binomial tests: $p < 0.0000001$, $p < 0.05$, $p < 0.00001$). Overall, monkey H selected $A_2 \rightarrow B_2$ on $88.9 \pm 13.9\%$ of trials across the three sessions (X^2 : $p < 0.001$). All three sessions for monkey T were also significant, selecting $A_2 \rightarrow B_2$ on 90%, 100%, and 96.7% of the trials (two-tailed binomial tests: $p < 0.00001$, $p < 0.0000001$, $p < 0.000001$). Overall, monkey T selected $A_2 \rightarrow B_2$ on $95.6 \pm 5.1\%$ of trials across the three sessions (X^2 : $p < 0.001$). All three sessions for monkey P were also significant and showed the opposite effect, selecting $A_2 \rightarrow B_2$ on 20%, 13.3%, and 26.7% of trials in three consecutive sessions (two-tailed binomial tests: $p < 0.01$, $p < 0.0001$, $p < 0.05$). Overall, monkey P selected $A_2 \rightarrow B_2$ on $20 \pm 6.7\%$ of trials across the three sessions (X^2 : $p < 0.001$). Thus, monkey P's preference was for $B_2 \rightarrow A_2$ over $A_2 \rightarrow B_2$.

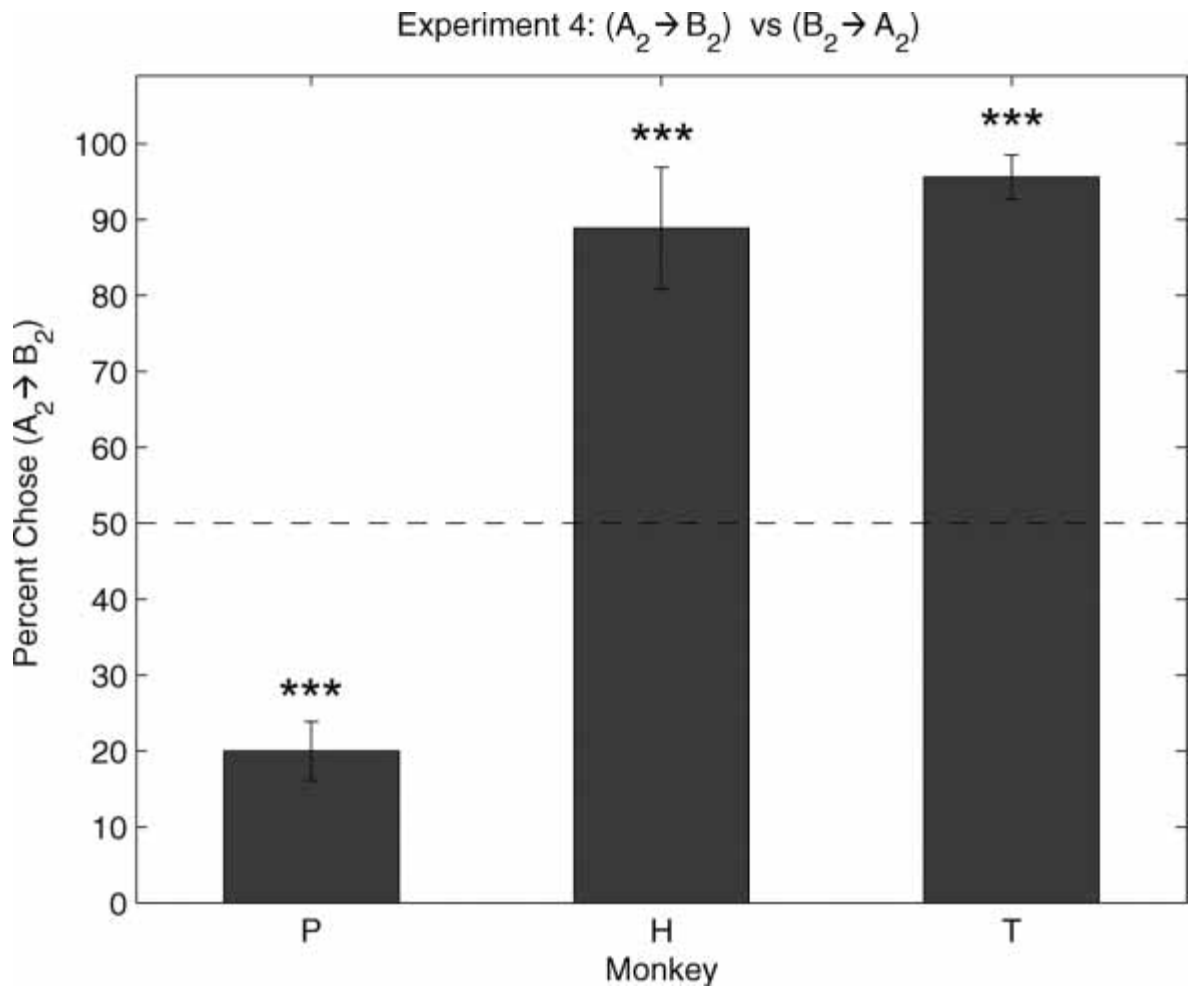


Figure 4. Reward order experiment (1A): Experiment 4 results, with one choice giving a mini marshmallow followed 4 s later by a Cheerio ($A_2 \rightarrow B_2$), and the other giving a Cheerio followed 4 s later by a mini marshmallow ($B_2 \rightarrow A_2$). The y axis plots the mean percentage of trials that each monkey selected ($A_2 \rightarrow B_2$) from the first significant session to the third significant session. Error bars are standard errors of measurement, the dashed line at 50% is random performance, and asterisks represent each monkey's chi-squared (X^2) p-value across the sessions: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

1A.6.4 Discussion

Overall, all three monkeys showed clear preferences. Monkey H and T once again exhibited a potential peak-first preference, suggesting that either (a) their choices were being influenced by only the first item received (thus, simply preferring A over B); or (b) receiving the best item first may be typically preferred by them, at least under these testing conditions. At the same time, with the two new food items, monkey P in fact showed a clear peak-end preference, which we find quite interesting. There was no evidence that a cheerio was ever valued more highly than a marshmallow for any of the monkeys (we in fact conducted three to four trials of preference testing on monkey P between trials 15 and 16—that is, in the middle of the session, during each of his three significant sessions, and he continued to select the marshmallow on every trial). Thus this finding appears to reflect a true peak-end preference for this monkey. The result therefore suggests that not all monkeys are compelled to obtain the best item first and/or immediately, and that the actual preferred order depends on the food items, at least for some monkeys.

Two possible factors underlying this preference reversal for monkey P are flavor–flavor order preferences and the differential value between the two food items. First, as in Experiment 3, it was possible that the consumption of the first item affected the actual taste of the second one, with the Cheerio being preferred before the marshmallow. Second, the differential value between the two food items might have caused the preference reversal. Further binary food preference testing with monkey P found that grapes were the most favored among the four food items, followed by marshmallows, green beans, and Cheerios (two-tailed binomial tests; grape>marshmallow: $p<0.05$,

grape>cheerio: $p=0$, marshmallow>green bean: $p<0.05$, green bean>cheerio: $p<0.05$).

The difference in value between grapes and green beans might be much larger than that between marshmallows and Cheerios, causing a peak-first preference in the former case, but not in the latter. It would be important to determine in the future whether the relative differences between the two food items can reliably predict a peak-first or peak-end preference. Flaherty, Turovsky, and Krauss (1994) did find that the effect of the second sweetened solution on the licking response of rats to the first solution depended on the relative value of the two rewards; however, their results appeared to be the opposite of ours. In their study, when the first reward was significantly lower than the second, a facilitation effect occurred on the first, whereas, as the reward value of the first increased and became closer to the level of the second, a negative contrast occurred. We, however, appeared to obtain the opposite effect with monkey P, such that when the first item was significantly lower than the second, the negative effect of the second item on the first increased. Further research will need to characterize the effects of the relative value of the two items on choice behavior.

1A.7. EXPERIMENT 5

In Experiment 1, all three monkeys exhibited a peak-first preference; however, it was not yet clear whether that preference was due to order per se, or whether it was driven by an impulsive tendency to receive the best item as soon as possible. Put differently, delay discounting theory would predict a preference for $A \rightarrow B$ over $B \rightarrow A$, because the higher value A would be significantly reduced when delayed (Ainslie, 1975; Logue, 1995; Rachlin & Green, 1972). We therefore conducted our final experiment to test for

the influence of order itself, independent of potential effects due to delay discounting. For this test, we again offered the choice between $A \rightarrow B$ and $B \rightarrow A$, but with a delay added before receiving the $A \rightarrow B$ outcome, so that the delay to receiving A was identical in both alternatives. Delay discounting theory should now predict a preference for the $B \rightarrow A$ alternative, since B is received immediately, and A is received at the same time as the identical first item A in the alternative. If, in contrast, the monkeys have a true order preference that is separable from the effects of delay discounting, then even with the added delay, $A \rightarrow B$ would still be preferred.

1A.7.1 Method

Subjects, materials, and procedure

These were as listed under “General method”.

1A.7.2 Specific procedure

Selection of one colored lid resulted in the monkey receiving a half vegetable, followed 4 s later by a grape, which was then followed by a 4-s delay, denoted $B \rightarrow A \rightarrow []$. The final delay was necessary to ensure a comparable trial length for both contingencies. The second colored lid gave an initial 5-s delay to match the exact overall delay to receive the grape in the first choice alternative, followed by a grape, followed 4 s later by a half vegetable, denoted $[] \rightarrow A \rightarrow B$. Table 5 shows lid colors and contingencies for each monkey.

Monkey	Canister colors/contingency	
	$B \rightarrow A \rightarrow []$	$[] \rightarrow A \rightarrow B$
P	Purple	Yellow
H	Green	Blue
T	Yellow	Striped (green and white)

Table 5. Reward order experiment (1A): The canister lid contingencies for Experiment 5.

Note: Selection of one canister resulted in a half vegetable, followed 4 s later by a grape, followed by a 4-s delay ($B \rightarrow A \rightarrow []$), and selection of the other resulted in an initial 5-s delay, followed by a grape, followed 4 s later by a half vegetable ($[] \rightarrow A \rightarrow B$).

1A.7.3 Results

As shown in Figure 5, two of the three monkeys (P and H) revealed a significant preference for $[] \rightarrow A \rightarrow B$ over $B \rightarrow A \rightarrow []$. All 3 sessions for monkey P were significant, selecting $[] \rightarrow A \rightarrow B$ on 86.7%, 90%, and 73.3% of the trials (two-tailed binomial tests: $p < 0.0001$, $p < 0.00005$, $p < 0.05$). Overall, monkey P selected $[] \rightarrow A \rightarrow B$ on $83.3 \pm 8.8\%$ of trials across the 3 sessions (X^2 : $p < 0.001$). Monkey H took 4 sessions to have 3 significant ones. In 3 consecutive sessions, monkey H selected $[] \rightarrow A \rightarrow B$ on 73.3%, 76.7%, and 76.7% of the trials (two-tailed binomial tests: $p < 0.05$, $p < 0.01$, $p < 0.01$). Overall, monkey H selected $[] \rightarrow A \rightarrow B$ on $75.6 \pm 1.9\%$ of trials across the last 3 sessions (X^2 : $p < 0.01$). Because monkey T did not have a significant session in 10 sessions, we concluded that he did not have a preference. Monkey T selected $[] \rightarrow A \rightarrow B$ on average $48.9 \pm 9.6\%$ of trials over his last 3 nonsignificant sessions (X^2 : $p > 0.2$).

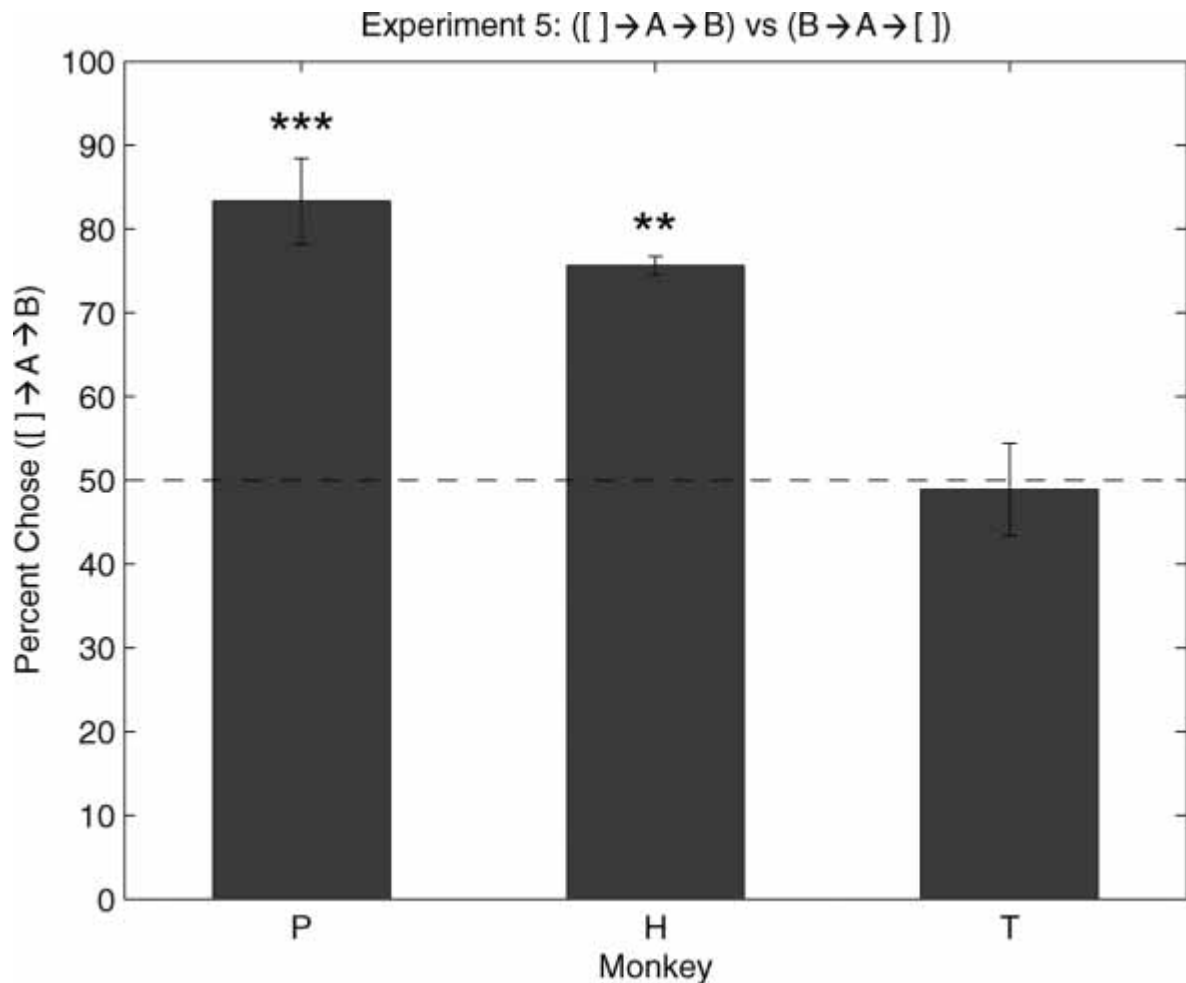


Figure 5. Reward order experiment (1A): Experiment 5 results, with one choice giving an initial delay of 5 s, followed by a grape, followed 4 s later by a half vegetable ([]→A→B), and the other giving a half vegetable, followed 4 s later by a grape, followed by another 4-s delay (B→A→[]). The y axis plots the mean percentage of trials that each monkey selected ([]→A→B) in the three significant sessions. Error bars are standard errors of measurement, the dashed line at 50% is random performance, and asterisks represent each monkey's significant chi-squared (X^2) p-value across the sessions: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Monkey T's result is the average and standard error of measurement over the final three nonsignificant sessions.

1A.7.4 Discussion

If the monkeys' choices had been based on how quickly they could obtain the rewards, they would have preferred $B \rightarrow A \rightarrow []$, because the overall outcome was delivered sooner than $[] \rightarrow A \rightarrow B$. This was not the case for any of the monkeys. Two of them (P and H) showed a clear preference for $[] \rightarrow A \rightarrow B$. Although it is possible that the delay in the last position could have had a stronger, negative influence on choice than that in the first position, causing the two monkeys to avoid $B \rightarrow A \rightarrow []$, it seems unlikely since Experiment 1 showed that the first position was more important than the last. Thus, monkeys P and H again exhibited a clear peak-first preference even when the items were received later in time. These results cannot be fully explained by delay discounting or risk aversion, and they provide evidence that order itself influenced the choices of these two monkeys.

Monkey T's indifference to the two alternatives suggests that the value of both Bs were equivalent for him, being either zero (or at least equal to the delay values) or some positive value. Because monkey T's results in Experiment 3 showed that the value of B in $B \rightarrow A$ was positive, they suggest that both Bs may be equally positive. Thus, for monkey T, it appeared that when A was immediate, it controlled his behavior, such that B was in fact ignored (Experiment 2). However, when A was delayed (both Experiments 3 and 5), monkey T was then influenced by B, when in either first or last position. His indifference between $B \rightarrow A \rightarrow []$ and $[] \rightarrow A \rightarrow B$, however, makes it unclear whether he was influenced by order, other than being exceptionally impulsive with the higher valued food item.

1A.8. GENERAL DISCUSSION

People appear to show a general peak-end preference, and we tested rhesus monkeys to see whether they showed an order preference. Based on our limited sample of 3 individuals, we found no evidence for a consistent peak-end preference. Most often, rhesus monkeys chose the option that provided the greatest good first: a peak-first preference. In a series of tests that explored this finding, we could rule out a simple account of our result in terms of delay discounting (Experiment 5), a consistent neglect of the lesser good (Experiment 2 and 3), or a specific relationship between the two goods tested in the main experiment (Experiment 4). Such factors seem to contribute to individual variation. One monkey (T), for example, neglected the lesser good when it was presented after a greater good and was indifferent to order when the delay of the greater good was the same. Another monkey (P) preferred the greater good first for familiar goods, but had the opposite preference for the more novel goods. These factors produce individual variation in the experiments, but they do not account for the main result: the lack of a consistent peak-end preference. These findings suggest that humans and rhesus monkeys value sequences of goods differently and point potentially to the importance of subsequent adaptations in food order preferences after apes diverged from other catarrhine primates roughly 25 million years ago.

Although a species difference may exist between humans and rhesus monkeys for valuation of sequential rewards, more studies are necessary both with humans and with nonhuman animals to clarify potential differences. As opposed to the Do et al. (2008) experiments, our study was conducted with the same subjects over repeated trials. It is

possible that the human findings would change if they were given more experience with the outcomes and tested in a repeated-choice paradigm (Hayden & Platt, 2009). As discussed above, there is evidence that forecasted preferences do not always match up with the subjective experience of the outcomes (Wilson & Gilbert, 2005), and preferences may change depending on particular test conditions, such as whether the alternatives are considered in isolation or in direct comparison (Christopher K Hsee, 1998; Christopher K Hsee, et al., 1999; Christopher K Hsee & Zhang, 2004; List, 2002; Mantonakis, Rodero, Lesschaeve, & Hastie, 2009; Rode, Rozin, & Durlach, 2007). Nonetheless, the peak-end phenomenon has been obtained with humans using a choice paradigm, thus resembling the present experiment (D. Kahneman, et al., 1993; Redelmeier, et al., 2003). In a recent experiment that measured the order people ate differently valued sushi (Jeong, et al., 2014), two significant populations were found. These significant populations included those showing a peak-end preference, preferring to eat the most-valued sushi last, and those showing a peak-first preference, preferring to eat the most valued sushi first. Thus although a peak-end effect is generally observed in people, humans appear to separate into distinct populations. In any case, it is clear that future experiments will need to test people and a diversity of species under more similar conditions to determine exactly how and when peak-end effects arise.

A peak-first preference appears to be similar to other phenomena found with nonhuman animals, such as relative impulsiveness and contrast effects in which the value of the food items influence each other. Indeed, monkey T in our study appeared to be exceptionally impulsive, such that when the higher valued food item (A) was received immediately, the other item appeared to be ignored, mimicking the selective-value effect

obtained with other primates (Beran, et al., 2009; Silberberg, et al., 1998). Interestingly, when A was delayed in Experiments 3 and 5, B then influenced monkey T's behavior because he preferred $B \rightarrow A$ over $[] \rightarrow A$ and was indifferent to $B \rightarrow A \rightarrow []$ and $[] \rightarrow A \rightarrow B$, respectively. This strong preference for receiving A immediately relates to findings that animals tend to prefer immediate over future rewards by devaluing or discounting the future goods (Ainslie, 1975; Hayden & Platt, 2007; Sarah R Heilbrunner, Rosati, Stevens, Hare, & Hauser, 2008; Rachlin & Green, 1972; Rosati, Stevens, Hare, & Hauser, 2007; Stevens, Hallinan, & Hauser, 2005).

Although monkey T's results can be mostly explained by general impulsiveness for the higher valued food item, the results of all three monkeys in Experiment 5 cannot be fully explained by delay discounting because none of the monkeys showed a preference for the alternative that gave both reward items earliest in time ($B \rightarrow A \rightarrow []$). Thus, apart from delay discounting, the order of the outcomes influenced choice.

The results in our study do not appear to be the result of memory limitations that might cause a primacy or recency effect, whereby the first (primacy) or last (recency) items in a serial list might be more easily remembered than others (Castro & Larsen, 1992; Wright, Santiago, & Sands, 1984; Wright, Santiago, Sands, Kendrick, & Cook, 1985). Wright and colleagues found that when the delay period was between 1–10 s, rhesus monkeys showed comparable primacy and recency on memory (Wright, et al., 1984; Wright, et al., 1985). Since our intertrial intervals were approximately 8 s and never more than 10 s, neither effect should have outweighed the other. Thus, our results do not appear to be an artifact of memory limitations. Blanchard and colleagues (Blanchard, et al., 2014) recently found that rhesus monkeys can show peak-end order

preferences in a *reward repeat task* for juice reward. The study's *reward repeat task* used a delay (intertrial interval) of 1 second, falling close to borderline where memory limitations may favor recency effects and promote a peak-end preference. Future work should examine the influence of intertrial interval between trials on choice order preferences.

Reward-value manipulation may also affect when order biases arise. Blanchard and colleagues' rhesus monkey study (Blanchard, et al., 2014) found a peak-end preference when manipulating value through quantity. However, Experiment 1 results in this study along with Jung and Kralik's study (Jung & Kralik, 2013) found a peak-first preference when reward quality was manipulated. Thus, another possible explanation for these differences is that reward quality can increase the relative value difference between outcomes and is more saliently remembered than reward quantity differences. This may ultimately promote enhanced primacy effects and increased delay discounting of the highly valued reward-outcome. Indeed the discrepant results for monkey P in Experiment 4 as opposed to Experiment 1 (see **1A.6.4 Experiment 4 Discussion**) may have resulted from a reduced relative value difference between marshmallows and cheerios, promoting a peak-end effect.

Task parameters such as whether outcomes are self-generated versus experimenter-generated could also affect when order biases arise. In Blanchard and colleagues' study (Blanchard, et al., 2014), the monkeys experienced an experimenter-generated passive experience of five juice squirts in quick succession and found a peak-end preference. In this case, the monkeys could be viewing the whole experience as one event, enhancing retrospective evaluation to guide future choice. Jung and Kralik's study

(Jung & Kralik, 2013), used an active self-generated choice task whereby the monkey picked on each of four trials which item it preferred from the reward-items available and found a peak-first preference. In this case, the monkeys likely viewed each of the four choices as separate events, enhancing prospective evaluation to maximize immediate reward. Thus it may be the case that Blanchard and colleagues' study (Blanchard, et al., 2014) enhanced the monkeys' retrospective evaluation of their choices promoting peak-end, whereas Jung and Kralik's study (Jung & Kralik, 2013) enhanced the prospective evaluation of their choices enhancing peak-first. In the current study, the monkeys received experimenter-generated sequences of only one or two reward items with a significant delay between them. These parameters, although experimenter-generated, may have made the outcome experience less likely to be viewed as a single event. Altogether, this distinction in task parameters can also account for the discrepant findings and our mixed results.

In general, contrast effects could underlie order preferences (see (Flaherty, 1999; Williams, 1997, 2002; Zentall & Singer, 2007)). For example, a negative anticipatory effect of A on B in $B \rightarrow A$ may lower the value of B, resulting in the lowering of the value of the entire alternative; likewise, a positive anticipatory effect (also called facilitation or positive induction) of B on A in $A \rightarrow B$ may increase the value of A and perhaps the entire alternative, resulting in a preference for $[] \rightarrow A \rightarrow B$ over $B \rightarrow A \rightarrow []$. However, because anticipatory contrast effects are typically obtained with different testing paradigms, dependent measures, and species, more work will be necessary to determine the degree to which contrast effects underlie order preferences in rhesus monkeys.

The order effects found here may be due to a similar phenomenon found in people

in which we often appear to neglect duration in our retrospective evaluations (Fredrickson & Kahneman, 1993; D. Kahneman, et al., 1993). Our monkeys may have also tended to neglect time as a factor, and instead focused more attention on the quality of the items and preferring the sequence with the best item first. People, on the other hand, may tend to neglect time for a different reason: when the experience results in a better end (peak-end).

Evolutionarily, it might be advantageous to follow a strategy such as “a fruit in the hand is better than two in the bush” when faced with the uncertainties of a dynamically changing and competitive environment (Sarah R Heilbrunner, et al., 2008; Rosati, et al., 2007; Stevens, et al., 2005). This logic would predict that if a human peak-end preference is at least partly an evolved adaptation, it would have developed as ecological resources for humans became more certain and predictable. However, more research is needed to clarify the differences between humans and other animals in decision making given that other studies have found similarities in choice behavior under uncertainty (Hayden & Platt, 2007, 2009; Hertwig, Barron, Weber, & Erev, 2004; Elke U Weber, Shafir, & Blais, 2004).

Overall, we have shown that the sequential ordering of reward items influences choice behavior. Future work should continue to determine the conditions under which factors such as reward order, risk, and delay discounting differentially influence decision making.

1B. The Effects of Reward Risk and Expected Value

Our lives are filled with choices, and many choices involve an element of risk. In principle, people should prefer options with the highest expected value, but in practice, different choices often prevail. Expressed in economic terms, people do not consistently maximize utility. The reason appears to involve the difficulty in computing utility, especially in natural conditions, when options are many and the outcomes uncertain or risky. We broadly define an option's risk level as the variance of its potential reward outcomes (Hayden, Heilbrunner, Nair, & Platt, 2008; McCoy & Platt, 2005; O'Neill & Schultz, 2010; Rothschild & Stiglitz, 1970; Elke U Weber, et al., 2004) or the square of the standard deviation of outcomes. As such, a risky option with more than one potential outcome has a greater risk value than that of a certain safe option. When an individual is given a choice between a safe and a risky option, there is an inherent opportunity cost (a form of expectation loss) based on the missed opportunities of the unchosen option(s).

Consider the calculations required to compare a certain and safe option to an uncertain and risky one. A safe option could involve receiving \$1 after every choice. A risky option with the same expected value could involve receiving \$2, but only after half of the choices, with the other half yielding nothing. The expected utility of the risky option is a function of the potential outcome values and their likelihoods: $(\$2 \times 0.5) + (\$0 \times 0.5) = \$1$. It is easy to see how such calculations could become computationally demanding with, for e.g., multiple choice options, each with multiple possible outcomes with different likelihoods.

Because of this computational complexity, our decision-making mechanisms have evolved to manage it using heuristics that substitute computationally difficult calculations

with approximations (Chen, Lakshminarayanan, & Santos, 2006; D. Kahneman, 2011; D. Kahneman & Frederick, 2002; D. Kahneman, Slovic, & Tversky, 1982; Jerald D. Kralik, Xu, Knight, Khan, & Levine, 2012; Lakshminarayanan, Chen, & Santos, 2011; Tversky & Kahneman, 1974, 1981). To understand how we make decisions, therefore, it is not sufficient to evaluate precise outcomes; we must also understand the operations of the cognitive processes underlying these heuristics.

In general, people are risk averse and avoid options that may result in a perceived loss of a “lesser” gain, no gain at all, or an actual loss. In addition, when outcome values and likelihoods are difficult to integrate, people tend to focus on specific components, such as the potential perceived losses or missed opportunities (opportunity costs), while ignoring others, such as likelihoods. Because of such factors, how problems are posed affects decision-making (D. Kahneman, 2011; D. Kahneman & Tversky, 1984; Tversky & Kahneman, 1974, 1981). For example, people’s preferences are influenced by whether the outcome contingencies are presented to them in oral or written form, as opposed to those discerned through experience. Choices are also affected by the number of decisions to be made, such as one opportunity to gamble vs. multiple opportunities (D. Kahneman, 2011; Klos, 2005; Montgomery & Adelbratt, 1982; Rabin & Vayanos, 2010; Sundali & Croson, 2006). In one-shot gambles, people appear to be particularly averse to the potential perceived loss and thus tend to select the safer option. With repeated gambles, people take on more risk, believing that the potential for possible perceived loss is mitigated as actual outcomes converge to the expected utilities (which reflects another bias, known as the Samuelson fallacy; (D. Kahneman, 2011; Klos, 2005; Montgomery & Adelbratt, 1982; Rabin & Vayanos, 2010; Samuelson, 1963; Sundali & Croson, 2006)).

Here we focus on experiential problems (as opposed to oral or written ones), which characterizes many problems faced by humans and all of those faced by other animals. Such problems are less susceptible to certain biases, but can still be difficult to evaluate. We also take a comparative perspective, which can provide some insight into the evolutionary history of decision-making processes and can also clarify possible advantages conferred by different types of risk and value assessments under different environmental conditions, independent of language-related and sociocultural factors. I use “risk preference” or “risk-seeking” for a preference for risky over safe options, and “risk aversion” or “safe(ity) preference” for *vice versa*.

There is evidence for a risk preference in rhesus monkeys (*Macaca mulatta*), a representative catarrhine monkey and a descendant of the last common ancestor of Old World monkeys, apes, and humans. For choices between options of equal overall expected utility, rhesus monkeys have been shown to have a preference for risky options (Hayden, et al., 2008; Hayden & Platt, 2007; McCoy & Platt, 2005; O'Neill & Schultz, 2010). For example, the monkeys preferred a risky option with a 50/50 chance of either 250 or 50 microliters of fruit juice to a safe option that offered 150 microliters 100% of the time. Recently, however, the generality of this conclusion has been questioned. Heilbronner and Hayden (S. R. Heilbronner & Hayden, 2013) suggested that the risk preference of rhesus monkeys might be context dependent, resulting from specific task parameters that might promote risk-seeking. These parameters include small differences in fluid reward, repeated gambles across hundreds, if not thousands, of trials, and short intertrial intervals of a few seconds or less. In fact, a recent study (Yamada, Tymula, Louie, & Glimcher, 2013) found rhesus monkeys to be mildly risk averse (i.e., the

percent of risky choices was less than 50%, but not significantly so). In this study, rhesus monkeys chose between a certain option that paid off a fixed amount of juice 100% of the time versus a risky option that paid off a particular amount 50% of the time, and nothing the remaining 50% of the time. Four conditions were used in which the amount of the safe option was changed. Within each condition five different risky options were used, each of different amounts, and thus the expected utility varied among these five risky options. In only the middle risky option in each condition were the expected utilities of the safe and risky options equal.

It remains unclear exactly why the risk preferences were different in this study versus the others that found a risky preference. Possible explanations include (1) higher amounts of juice used; (2) non-rewarded trials on 50% of risky choices; (3) significant experience with a fixed probability structure for the risky options and multiple different values that could have trained the monkeys to use both probability and value and thus expected utility more accurately; and (4) pie charts displayed at the beginning of the trial to represent value that may have helped simplify (e.g., linearize) the value estimation.

1Bi. Food-Based Gambling Task

To help delineate the key factors influencing risky decision-making, we modified several of them while maintaining equal expected values between options for Experiments 1–6 as in the original studies (Hayden, et al., 2008; Hayden & Platt, 2007; McCoy & Platt, 2005; O'Neill & Schultz, 2010). After first determining the relative value of three qualitatively different food items, I conducted seven short experiments using them as reward.

Compared to the previous studies, I increased the stakes by using fewer trials per session, a longer intertrial interval, larger reward amounts, and greater variation in the risky outcomes. In addition, rather than fluid reward, our study used food items. The risky options also included non-rewarded outcomes similar to Yamada and colleagues' study (Yamada, et al., 2013). In the first six experiments, the monkeys preferred the risky option. Finally, I tested whether their risk preference was due to a focus on the highest outcome payoff of the risky option (a salience bias), with neglect of likelihoods. Both monkeys exhibited sensitivity to the likelihoods, reversing their preferences to the safe option when it paid off four times more than the risky one.

1Bi.1. MATERIALS AND METHODS

1Bi.1.1 Subjects

Two male rhesus monkeys were available for this study, denoted as Monkey T and P, ages 11 and 9, respectively. Animal care and use complied with all current laws, regulations, policies, and guidelines of the United States, the United States Department of Agriculture (USDA), the Public Health Service (PHS), and all procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of Dartmouth College.

The Center for Comparative Medicine and Research (CCMR) at Dartmouth maintains a full-time animal care and veterinary staff that monitors the monkeys' daily health and well-being.

The monkeys were housed in a homeroom with automatically regulated temperature, ventilation, humidity, and lighting. The monkeys were intermittently housed together and individually: at times when they engaged in confrontations, which is normal periodic behavior in young rhesus macaque males of similar size and temperament (Thierry, Mewa, & Kaumanns, 2004), the two monkeys were separated and individually housed for their safety. When pair-housed, they had direct physical contact with each other, and when individually housed, through a mesh grading divider between their cages. They also had direct visual contact with the other monkeys in the colony, as well as the animal care staff and experimenters. In addition, environmental enrichment included two or more enrichment items in their home cages at all times, daily playing of radio or videos in the room (the latter via a monitor mounted in view of all individuals), and regular access to a larger enrichment cage (172.7 cm × 96.5 cm × 182.9 cm) in an adjacent room. The monkeys had *ad libitum* access to food and water during the study.

1Bi.1.2 Materials and testing procedures

In the test room, the monkey's chair was placed on the opposite side of a table, 76.2 cm, from the experimenter. A monkey sat in the chair, with his left arm loosely restrained (using two custom-made metal rings attached to the chair around both the upper and lower arms) and his right arm free to reach.

In food preference testing, the monkeys were presented with two rectangular compartments (each 7.62cm × 8.89 cm), which had transparent lids in order to see the

food items directly. In Experiments 1–7, which tested risk preferences, the monkeys were presented with one or two opaque compartments, also 7.62 cm × 8.89 cm, which might or might not contain food items. For these seven experiments, the left compartment lid was blue with a green center cross, and the right compartment lid was green with a blue center cross (see **Figure 6**).

Each compartment was glued to its own underlying Plexiglas® sheet (12.7 cm × 35.56 cm) with the compartment at the monkey's end of the sheet. The compartments could thus be moved both simultaneously (with the sheets touching on adjacent edges) and independently. An opaque plastic divider separated the experimenter and the monkey. An opening at the bottom of the divider allowed the experimenter to present both compartments simultaneously by placing each hand on the back of a Plexiglas sheet and sliding the compartments toward the monkey. The monkeys made selections by lifting a compartment lid to obtain the contents inside.

To minimize the potential for inadvertent cues from the experimenter, several procedures were implemented: (a) as just mentioned, an opaque plastic divider separated the experimenter and the monkey, thus preventing the monkey from seeing the face and upper body of the experimenter; (b) white noise was played to minimize auditory distractions; and (c) the experimenter followed a well-practiced routine on each trial.

For all testing in the study (food preference tests and experiments), each trial began with the experimenter holding the Plexiglas apparatus so that both compartments were 40.6 cm in front of the monkey but not within the monkey's reach. The experimenter held this position until the monkey looked at the compartments, as determined by monitoring the monkey via a close-circuit camera and video monitor.

Next, both compartments were pushed forward to a fixed position within the monkey's reach. Once the monkey touched the lid of one of the compartments, the unselected compartment was quickly removed. The intertrial interval throughout the study was approximately 15 s.

To rule out potential side biasing effects in food preference testing, the food item positions (left or right compartment) were randomized within a daily session, with the constraint that each food item was presented to the left and right an equal number of times in the session. For each monkey, 60 trials were conducted per daily session, and two sessions were conducted for each of the three food preference testing conditions.

For risk preference testing (Experiments 1–7), I conducted the trials in blocks, with each block consisting of 16 trials: eight forced choice trials—four forced choice trials from one of the two compartments (randomly determined) followed by four forced choices from the other compartment—and eight free choice trials (**Figure 6**). The monkeys typically performed four blocks per daily session with 12 blocks per experiment (further detail below). The experiments were conducted in consecutive order and only one experiment was conducted in any daily session (thus, on sessions in which an experiment was completed, the session terminated, and the new experiment was begun the following session). The experiment the monkey experienced in each daily session was not cued; rather, I used the forced choice trials conducted at the beginning of every block as the means by which the monkeys learned the contingencies of the given experiment. Table 6 summarizes the outcome contingencies for both the risky and safe options for each experiment, along with (a) the variance of the risky option outcomes as a measure of risk and (b) the utility/expected value, i.e., the sum of the products of value by

likelihood for all option outcomes, of the risky options.

To minimize the potential for the actual experienced payoffs associated with the risky option deviating significantly from the assigned contingencies, the outcomes of the risky options were set up in the following way. I first describe the free choice trials. In all experiments, the outcomes associated with the risky option occurred according to the assigned proportions across each block of eight consecutive free choice trials. Thus, for example, in Experiment 1 four of the eight free choice trials were pseudo-randomly assigned to pay off two mini marshmallows; and thus every free choice block of eight trials paid off exactly as listed in Table 6. I next describe the forced choice trials. For all experiments except for 5 and 7, the outcomes associated with the risky option occurred according to the assigned proportions across each block of four consecutive forced choice trials. Thus, for example, in Experiment 1 two of the four forced choice trials were pseudo-randomly assigned to payoff two mini marshmallows. For Experiments 5 and 7, the outcomes associated with the risky option occurred according to the assigned proportions across every two blocks, and thus across every eight forced choice trials. Thus, for example, in Experiment 5 one of the eight forced choice trials across every two blocks (i.e., four in the first block, four in the second) was pseudo-randomly assigned to payoff eight mini M&Ms. As reported in the section **1Bi.2.9.2 Potential response strategies**, I found no evidence that the monkeys were sensitive to the local changes in the trial-by-trial outcome probabilities due to this baiting procedure.

To rule out the potential influence of location biases, each experiment was conducted in two configurations in which the safe and risky option switched compartment positions. The compartments retained their same color and shape features,

and thus the monkeys had to discern the flipped contingencies based on the forced choice trials. In Experiment 1, Monkey P was run on 11 blocks in one configuration and 10 blocks in the other, while Monkey T was run on eight blocks in each configuration. In Experiment 2, Monkey P was run on eight blocks in one configuration and six blocks in the other, while Monkey T was run on 12 blocks in one configuration and 11 blocks in the other. In Experiments 3–7, both monkeys were run on 12 blocks in each configuration. To obtain the monkeys' stable choice preference in each experiment, I analyzed the last 30 free choice trials in both configurations combined, i.e., 60 total trials per experiment. All binomial tests were two-tailed.

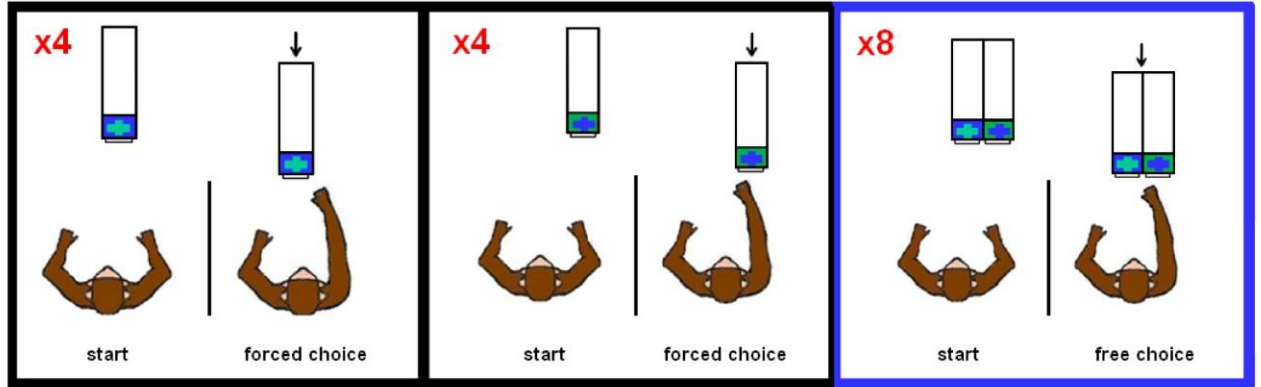


Figure 6. Food-Based Gambling Task (1Bi): Task design schematic of Experiments 1-7.

Each block consisted of four forced choice trials on each side (order randomly determined) followed by eight free choice trials. For each trial, the Plexiglas apparatus was presented to the monkey out of reaching distance before it was moved forward for selection.

Experiment	Safe Option	Risky Option		
		Contingencies	Variance	Utility / Expected Value
1	1B (100%)	2B (50%) & 0 (50%)	1	1B
2	1A (100%)	2A (50%) & 0 (50%)	1	1A
3	1A (100%)	4A (25%) & 0 (75%)	2.99	1A
4	1A (100%)	2A (50%) & 0 (50%)	1	1A
5	1A (100%)	8A (12.5%) & 0 (87.5%)	7.02	1A
6	1C (100%)	2C (50%) & 0 (50%)	1	1C
7	1A (100%)	2A (12.5%) & 0 (87.5%)	0.44	0.25A

Table 6. Food-Based Gambling Task (1Bi): The reward contingencies in the seven experiments (A: mini M&M, B: mini marshmallow, and C: cheerio). Utility/Expected Value: the expected value of the risky option per trial by objective calculations.

1Bi.2. RESULTS

In a series of experiments in which the mean utilities between the safe and risky options were equivalent, I manipulated the value of the gamble via changes in reward quality (Experiments 1, 2, 4, 6), the risk level via changes in the variance of risky choice outcomes (Experiments 2–5), and novelty (Experiment 4 replicating Experiment 2). Despite these changes, both monkeys continued to prefer the risky over the safe option. Only when the utility associated with the safe option was greater than that for the risky option did the monkeys prefer the safe option (Experiment 7).

1Bi.2.1 FOOD PREFERENCE

I first conducted food preferences to assess the relative value of food items that differed in quality. I tested three food items using a two alternative free choice procedure: a mini M&M® (food A), a mini marshmallow (food B), and a cheerio® (food C). The monkeys were familiar with these food items, but only received them during the study in the food preference tests and experiments.

1Bi.2.1.1 Food preference results

For mini M&Ms (A) vs. cheerios (C), both monkeys exhibited a significant preference for the former (Monkey P, 99.2%, binomial test, $z = 10.68$, $p < 0.001$; Monkey T, 90.8%, binomial test, $z = 8.85$, $p < 0.001$). They also exhibited a significant preference for mini M&Ms (A) over mini marshmallows (B) (Monkey P, 64.2%, binomial test, $z = 3.01$, $p < 0.01$; Monkey T, 73.3%, binomial test, $z = 5.02$, $p < 0.001$) and for mini marshmallows

(B) over cheerios (C) (Monkey P, 90.8%, binomial test, $z = 8.85$, $p < 0.01$; Monkey T, 80.0%, binomial test, $z = 6.48$, $p < 0.001$).

1Bi.2.1.2 Food preference discussion

Overall, both monkeys exhibited a clear transitive preference among the three food items ($A > B > C$). They valued mini M&Ms the most, mini marshmallows second, and cheerios least. Neither the food's mass, caloric value, nor sugar content fully accounted for these preferences. One mini M&M had a mass of 0.30 g (0.19 g sugar) and 1.47 cal; one mini marshmallow had 0.50 g (0.28 g sugar) and 1.67 cal; and one cheerio had 0.10 g (0.03 g sugar) and 0.39 cal. So the preferred subjective values did not fall in the same order as the objective values, although cheerios came out last in both. In terms of size (volume), the preferred food, mini M&Ms, was the smallest of the three foods. Thus, although mini marshmallows were the largest, heaviest, most caloric, and contained the most sugar, the monkeys preferred the mini M&Ms, perhaps because of its chocolate content or its higher sugar density [mini M&Ms (A): 1.06 g/mm^3 ; mini marshmallows (B): 0.56 g/mm^3 ; cheerios (C): 0.33 g/mm^3]. Finally, the food preference results also show that even though the monkeys were not food restricted, they were sufficiently motivated to select and consume the items, and they held clear preferences among the items.

1Bi.2.2 EXPERIMENT 1

All trials consisted of a safe option in one compartment and a risky option in the other. The safe option always contained one mini marshmallow, while the risky option yielded

two mini marshmallows 50% of the time and nothing the remaining 50% of the time.

The expected value of both options was 1 mini marshmallow (B).

1Bi.2.2.1 Experiment 1 results

Both monkeys exhibited a significant preference for the risky option over the safe option (**Figure 7**). Monkey P selected the risky option at an 86.7% rate (binomial test, $z = 5.55$, $p < 0.001$) and Monkey T did so at an 83.3% rate (binomial test, $z = 5.03$, $p < 0.001$).

1Bi.2.2.2 Experiment 1 discussion

Both monkeys preferred the risky option over the safe one when foraging for mini marshmallows, the intermediately valued food item. McCoy and Platt (McCoy & Platt, 2005) found a risk preference using time access to fluid rewards as their measure, and our results generalize this finding to discrete food quantities. Experiment 1 also differs from previous ones in that our monkeys exhibited a risk preference with relatively few trials and a longer ITI (Hayden, et al., 2008; Hayden & Platt, 2007; McCoy & Platt, 2005; O'Neill & Schultz, 2010), thus further generalizing the finding. We can only infer the subjective value of food items, although the monkeys appear to place a high value on the marshmallows. We cannot rule out the possibility that their motivation was insufficient to be affected by the “0” trials of the risky option, i.e., the trials on which the monkeys obtained no marshmallows. In Experiment 2, I used mini M&Ms, a more valued food item, to determine if increasing the intrinsic reward value would change the monkeys' risk preference.

1Bi.2.3 EXPERIMENT 2

Subjects, Materials, and Methodology were the same as in Experiment 1 except for the food used: mini M&Ms.

1Bi.2.3.1 Experiment 2 results

As Figure 7 shows, both monkeys exhibited a significant preference for the risky over the safe option. Monkey P and T selected the risky option 86.7% (binomial test, $z = 5.55$, $p < 0.001$) and 70.0% (binomial test, $z = 2.97$, $p < 0.01$) of the time, respectively.

1Bi.2.3.2 Experiment 2 discussion

The more valuable food item, mini M&M (A), did not alter the monkeys' general risk preference as compared to mini marshmallow (B), the intermediately valued food item used in Experiment 1. Thus, the intrinsic reward value difference between these two food items did not have a significant effect on the monkeys' general risk preference. I test reward quality again in Experiment 6.

1Bi.2.4 EXPERIMENT 3

The coefficient of variation (CV) is another well-known measure of risk that divides the standard deviation of an option's outcomes (square root of variance) by its mean utility (Elke U Weber, et al., 2004). Because the CV produces a measurement of relative risk, it allows for comparison between studies. In McCoy and Platt's (2005) study, the monkeys' risk preference increased with the CV of the risky option as it was progressively increased from 0.0667 to 0.667. In our Experiments 1 and 2, the CV (standard deviation/expected value) for the risky option was 1. It is unclear whether risky

behavior would continue when the risky option's CV becomes >1 . In addition, the likelihoods of the outcomes in the risky options used in previous monkey studies (Hayden, et al., 2008; Hayden & Platt, 2007; McCoy & Platt, 2005; O'Neill & Schultz, 2010) have been 50–50%. In such cases, the high outcome has a relatively high possibility per trial. In Experiment 3, I addressed these issues by increasing the risky option's CV above 1 and lowering the likelihood of the high reward outcome.

Subjects, Materials, and General Methodology were the same as in Experiment 1. All trials consisted of a safe option in one compartment and a risky option in the other. The safe option always contained one mini M&M, while the risky option yielded four mini M&Ms 25% of the time and nothing the remainder of the time (**Table 1**). The variance of the risky option was 2.99 ($CV=1.73$), and the expected value of both options was 1 mini M&M.

1Bi.2.4.1 Experiment 3 results

Both monkeys continued to exhibit a significant preference for the risky option over the safe option, and did so to a greater extent than in Experiments 1 and 2 (**Figure 2**).

Specifically, Monkey P and T selected the risky option 100% (binomial test, $z = 7.62$, $p < 0.001$) and 96.7% (binomial test, $z = 7.1$, $p < 0.01$) of the time, respectively.

1Bi.2.4.2 Experiment 3 discussion

Both monkeys strongly preferred the risky option over the safe option with the new experimental contingencies. Hence, the risk preference persisted despite increasing the

CV of the risky option from 1 to 1.73, reducing the likelihood of the highest outcome from 50% to 25%, and increasing the highest outcome from two discrete mini M&Ms to four. The results suggest that the monkeys might be more attentive to the large reward outcome and/or the outcome's risk level, although further testing is needed to assess the generality of this conclusion.

1Bi.2.5 EXPERIMENT 4

The risk preference observed in Experiment 3 might have been due to the novelty of the risky option's new reward contingency compared to Experiment 2 (with the safe option remaining the same, but the risky option contingency changing). We do not believe this to be the case for two reasons: (1) my analysis used the last 30 trials in each configuration after sufficient learning and a stable preference was obtained; and (2) in Experiments 1 and 2 both the safe and risky option contingencies were novel, and yet the monkeys exhibited preferences for the risky option. However, we nonetheless addressed this issue in Experiment 4 by repeating Experiment 2 to test whether the risky option's novelty in Experiment 3 compared to Experiment 2 led to the risky choice preference in Experiment 3. Furthermore, to standardize the block sizes across configurations and subsequent experiments, I set the block size at 12 per configuration beginning in Experiment 3. I therefore also repeated Experiment 2 to test the contingencies under the standardized protocol to enable direct comparisons across experiments (see **1Bi.3.9.1 Effects of contingency parameters**).

The methods for Experiment 4 were the same as described in Experiment 2.

1Bi.2.5.1 Experiment 4 results

Both monkeys exhibited a significant preference for the risky option, as they had in Experiment 2 (**Figure 7**), at a rate of 100% for Monkey P (binomial test, $z = 7.62$, $p < 0.001$) and 93.3% for Monkey T (binomial test, $z = 6.58$, $p < 0.001$).

1Bi.2.5.2 Experiment 4 discussion

Both monkeys continued to prefer the risky over the safe option. Thus, the risk preference of the monkeys observed in Experiment 3 appeared to be a general preference rather than one based on the novelty of the risky option's contingency in Experiment 3 compared to Experiment 2. In addition, I replicated the general risky preference exhibited in Experiment 2.

1Bi.2.6 EXPERIMENT 5

Because the monkeys' risky option preferences did not seem to be affected by the novelty of the risky option's contingencies, I explored more extreme parameters. In Experiment 5, the safe option always contained one mini M&M, while the risky option yielded eight variance M&Ms 12.5% of the time and nothing the remainder of the time. Thus, the variance of the risky option was 7.02, and the mean expected value of both options was 1 mini M&M.

1Bi.2.6.1 Experiment 5 results

As in the previous experiments, both monkeys exhibited a significant preference for the risky option over the safe option (**Figure 7**), at a rate of 100% (binomial test, $z = 7.62$, p

< 0.001) for Monkey P and 93.3% (binomial test, $z = 6.58$, $p < 0.001$) for Monkey T, much like Experiments 3 and 4.

1Bi.2.6.2 Experiment 5 discussion

Despite the three changes from Experiment 3 – increasing the CV of the risky option from 1.73 to 2.65, reducing the likelihood of the risky option's positive outcome from 25 to 12.5%; and increasing the risky option's positive reward outcome from four discrete mini M&Ms to eight—the monkeys reliably preferred the risky option.

1Bi.2.7 EXPERIMENT 6

An important issue is the extent decision-making preferences change based on one's intrinsic motivation level. In principle, motivation relates to both the current state of the organism, i.e., their level of “wealth,” and the current state of the environment, whether rich or lean. Results thus far are mixed, with individuals appearing to become more and less risk-seeking with increasing motivation in different studies (Caraco, Martindale, & Whittam, 1980; e Abreu & Kacelnik, 1999; Kacelnik & Bateson, 1996, 1997; Yamada, et al., 2013). In Experiment 6, I tested whether a decrease in the motivation level might result in a preference reversal to the safe option by decreasing the intrinsic value of the food reward. To do so, I repeated the outcome contingencies of Experiments 1, 2, and 4 but replaced the food items with cheerios, the least preferred food.

The safe option always contained one cheerio, while the risky option yielded two cheerios 50% of the time and nothing the other 50% of the time (**Table 6**).

1Bi.2.7.1 Experiment 6 results

Monkey P and T selected the risky option 90% (binomial test, $z=6.07$, $p<0.001$) and 88.3% (binomial test, $z=5.81$, $p<0.001$) of the time, respectively (**Figure 2**).

1Bi.2.7.2 Experiment 6 discussion

Both monkeys preferred the risky over the safe option using cheerios as reward. Thus, a change in reward quality did not change the overall risk preferences of the monkeys, at least within the range tested in this study. However, evidence was found for some effect of intrinsic reward value (see **1Bi.3.9.1 Effects of contingency parameters**).

1Bi.2.8 EXPERIMENT 7

A possible account for the risk preference in Experiments 1–6 was that the monkeys might be driven toward the option with the highest possible reward outcome (a salience bias). Along with this possibility, the monkeys might also be relatively insensitive to the likelihood-based valuations of the options. This could be due to the stakes in which there is no objective loss other than opportunity costs.

To test these possibilities, in Experiment 7 the monkeys chose between a safe and a risky option with the risky option having a lower mean utility compared to the safe one. The risky option still had potential for the largest reward outcome in the experiment, as in Experiments 1–6.

The safe option always contained one mini M&M, while the risky option yielded two mini M&Ms 12.5% of the time and nothing the remainder of the time. Thus, the variance of the risky option was 0.44 and its mean expected value was 0.25 mini M&Ms.

As in Experiments 2–5, the mean expected value of the safe option was 1 mini M&M, four times larger than the risky option in this experiment (**Table 6**).

1Bi.2.8.1 Experiment 7 results

Unlike the results of Experiments 1–6, both monkeys exhibited a significant preference for the safe option over the risky one (**Figure 7**). Specifically, Monkey P selected the risky option at a rate of 8.3% (binomial test, $z = -6.33$, $p < 0.001$) and Monkey T did so at 21.7% (binomial test, $z = -4.26$, $p < 0.001$).

1Bi.2.8.3 Experiment 7 discussion

These results show that neither monkey simply chose the option that yielded the most valuable possible outcome, which would have reflected a salience bias. The results further show that both monkeys integrated likelihood and reward value across trials to make their decisions.

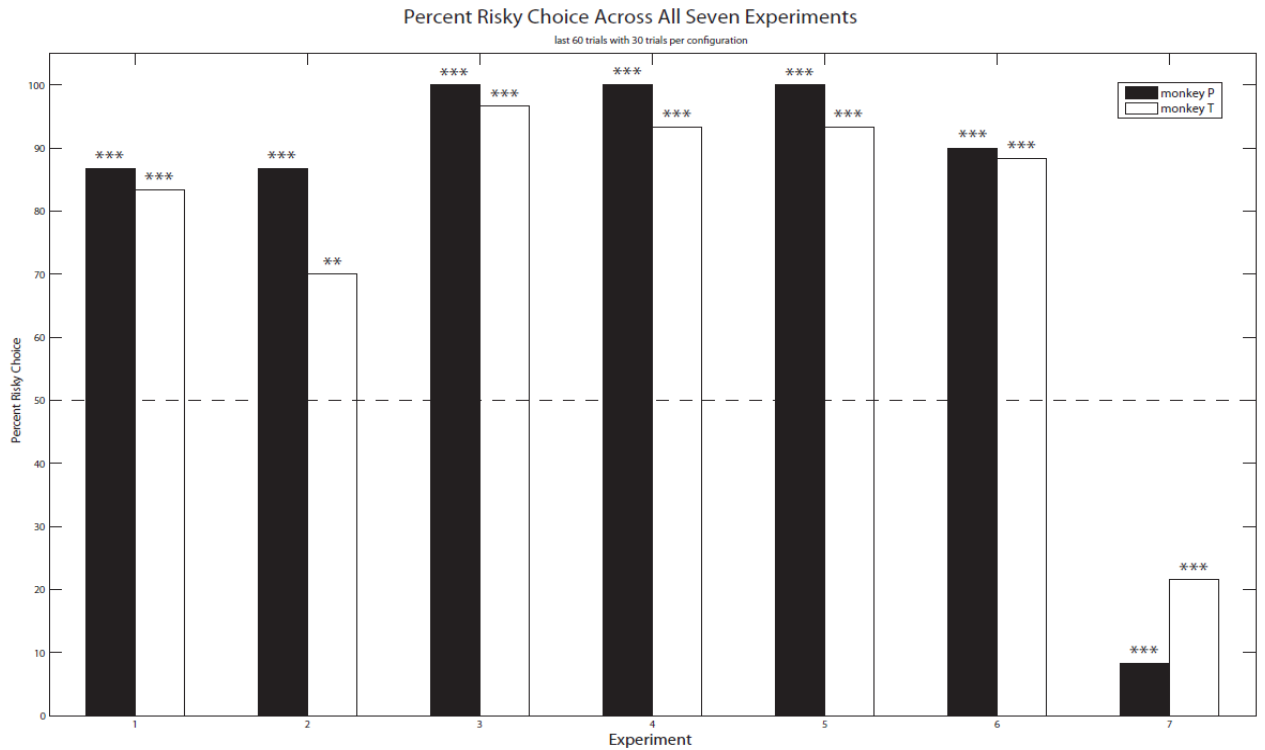


Figure 7. Food-Based Gambling Task (1Bi): Experiment 1–7: The mean percentage of trials on which each monkey selected the risky option over the last 30 free-choice trials in both configurations (60 total trials). Dashed line: no preference; Asterisks: significant preference (two-tailed binomial test, $**p < 0.01$, $***p < 0.001$).

1Bi.2.9 Comparison across experiments

1Bi.2.9.1 Effects of contingency parameters

Because Experiments 1 and 2 were conducted with fewer than 12 blocks per condition, I compared the percent of risky choices in Experiments 3-7. A comparison of Experiments 4 and 6 (**Table 6**) showed that there was an effect of reward quality with the percent risky choice decreasing with a decrease in reward quality in both monkeys; however, the decrease was significant only for Monkey P [$X^2(1) = 6.316, p = 0.012$; Monkey T, $X^2(1) = 0.901, p = 0.343$]. Thus, to the extent overall reward quality influences motivation, there was some evidence for greater motivation leading to an increased risk preference. A comparison of Experiments 3 and 4 [Monkey P, $X^2(1)=0, p=1$; Monkey T, $X^2(1)=0, p=1$], Experiments 4 and 5 [Monkey P, $X^2(1) = 0, p = 1$; Monkey T, $X^2(1)=0, p=1$], and Experiments 3 and 5 [Monkey P, $t(2)=3.28, p = 0.08$; Monkey T, $t(2) = 0.35, p = 0.76$] found no relationship between variance (our measure of risk) and the percent of risky choices for either monkey, which could have been due to a ceiling effect. Finally, a comparison of Experiments 4 and 7 showed that likelihood affected the percent of risky choices in both monkeys [Monkey P, $X^2(1) = 127.03, p < 0.001$; Monkey T, $X^2(1) = 60.92, p < 0.001$], and thus preferences could not be explained by a neglect of the non-rewarded outcomes of the risky options.

1Bi.2.9.2 Potential response strategies

Although there was some evidence for persistence or response habits when the configurations changed, nonetheless, in every experiment, both monkeys changed their choice to the risky (Experiments 1–6) or safe option (Experiment 7). Even so, strategies

such as “win-stay, lose-shift” or “win-shift, lose-stay” were possible (Genovesio, Brasted, Mitz, & Wise, 2005; S. R. Heilbronner & Hayden, 2013). Due to my baited procedure for determining the outcomes of the risky options (see **1Bi.1.2 MATERIALS AND TESTING PROCEDURES**), a “win-shift, lose-stay” strategy would be a sign that the monkeys were tracking the changes in the trial-by-trial probabilities instead of showing a true overall risk preference. However, I found no evidence for this response strategy by either monkey in any experiment. Furthermore, I found no evidence for the “win-stay, lose-shift” response strategy in any experiment except for one monkey in Experiment 7, when mean expected values between options were unequal. In all other cases, comparing the proportion of trials each monkey stayed with the risky option after receiving the largest outcome with the proportion each monkey stayed with the risky option when he did not receive the largest outcome, no significant differences were found [Experiment 1: Monkey P, $X^2(1) = 0.15$, $p = 0.70$; Monkey T, $X^2(1) = 0.08$, $p = 0.77$; Experiment 2: Monkey P, $X^2(1) = 0.02$, $p = 0.89$; Monkey T, $X^2(1) = 3.04$, $p = 0.08$; Experiment 3: Monkey P, $X^2(1) = 0.09$, $p = 0.77$; Monkey T, $X^2(1) = 0.75$, $p = 0.39$; Experiment 4: Monkey P, $X^2(1) = 1.94$, $p = 0.16$; Monkey T, $X^2(1) = 1.10$, $p = 0.29$; Experiment 5: Monkey P, $X^2(1) = 0.54$, $p = 0.46$; Monkey T, $X^2(1) = 3.04$, $p = 0.08$; Experiment 6: Monkey P, $X^2(1) = 0.41$, $p = 0.52$; Monkey T, $X^2(1) = 0.01$, $p = 0.91$; Experiment 7: Monkey P, $X^2(1) = 1.24$, $p = 0.276$]. The exception was Monkey T in Experiment 7, in which he appeared to exhibit a “win-stay, lose-shift” strategy with the risky option ($X^2(1) = 4.22$, $p = 0.04$). Thus, except for Monkey T in Experiment 7, the response strategy that best reflected their choice behavior was “win-stay, lose-stay,” suggesting that the monkeys were exhibiting an overall risk preference for Experiments 1–6.

In Experiments 5 and 7, in the forced choice trials, the outcomes associated with the risky option occurred according to the assigned proportions across every two blocks, and thus across every eight forced choice trials (i.e., four forced choice trials in one block, and four in the second). This baiting procedure resulted in trial blocks in which one of the risky forced choices was rewarded, followed by a trial block in which none of the risky forced choices were rewarded (or vice versa). If the monkeys were sensitive to the local changes in the trial-by-trial outcome probabilities due to this baiting procedure, there should be a difference in the percent of risky choices in the blocks in which the large payoff was received in the forced choice trials versus when no reward was received in the forced choice trials. To test for a potential pattern in Experiments 5 and 7, I examined the free choices in the last four blocks in each left–right configuration (eight blocks total). I compared the four blocks in which one of the four risky forced choices was rewarded to the four blocks in which none of the risky forced choices were rewarded. Neither monkey showed a significant behavioral difference between these blocks in either experiment [Experiment 5: Monkey P, $X^2(1) = 0$, $p = 1$; Monkey T, $X^2(1) = 0.35$, $p = 0.55$; Experiment 7: Monkey P, $X^2(1) = 0.22$, $p = 0.64$; Monkey T, $X^2(1) = 1.46$, $p = 0.23$], and thus they appeared to be following the general risky option contingency rather than local trial-by-trial changes.

1Bi.2.9.3 Dynamics across experiments

Table 7 shows the number of trials in which each monkey first exhibited his final preference in each left–right configuration (examined with a sliding window of 16 free choice trials). For example, in Experiment 1, Monkey P exhibited a significant risk preference after 75 free choice trials in his first experienced configuration, and then

showed a significant risk preference after 66 free choice trials when the safe and risky options changed positions. We found no significant decrease (or increase) across all experiments either overall or for either configuration for either monkey (linear regression using the values in each column of Table 7, $p > 0.05$), suggesting that the monkeys were learning each new contingency for each experiment.

Examining only the experiments with risk preferences, i.e., Experiments 1–6, neither monkey exhibited a significant change overall (linear regression, Monkey P: $R^2 = 0.63$, $p = 0.059$; Monkey T: $R^2 = 0.13$, $p = 0.833$), although Monkey P was trending toward significance ($p = 0.059$), which was due to a significant decrease for Configuration 1 ($R^2 = 0.8$, $p < 0.05$) but not Configuration 2 ($R^2 = 0.28$, $p = 0.28$). In addition, given identical reward contingencies in Experiments 2 and 4, a comparison of the acquisition rates to acquire a significant preference in the two experiments enabled a direct test of experience on choice behavior. The acquisition rates did not differ [Monkey P, $t(2) = 0.21$, $p = 0.85$; Monkey T, $t(2) = 1.10$, $p = 0.39$], providing evidence against an experiential effect across experiments. Finally, for both monkeys, the total number of trials to achieve significance in both configurations was the lowest (and thus the fastest acquisition rate) for Experiment 5, in which they could obtain the largest amount of the preferred food item on any risky trial—eight mini M&Ms— 12.5% of the time.

Experiment	Monkey P			Monkey T		
	Conf. 1	Conf. 2	Total	Conf. 1	Conf. 2	Total
1	75	66	141	63	22	85
2	48	27	75	64	84	148
3	42	43	85	20	71	91
4	44	25	69	56	67	123
5	21	31	52	33	40	73
6	28	37	65	77	35	112
7	69	78	147	78	31	109

Table 7. The number of trials in which each monkey first showed his final preference, with preference measured using a sliding window of 16 consecutive free choice trials, and a preference-significance threshold of $\geq 13/16$, ($p < 0.05$, two-tailed binomial test), assessed for each configuration separately. Conf.: configuration.

1Bi.3. DISCUSSION

1Bi.3.1 Risk preference: Experiments 1-6

The preferences for risky options reported thus far for macaque monkeys are potentially context driven, resulting from specific task parameters that might promote risk-seeking (S. R. Heilbronner & Hayden, 2013). Previous studies that found a risk preference in rhesus monkeys have used small reward amounts and differences, repeated gambles across hundreds of trials, and short intertrial intervals, all of which may promote risk-seeking behavior (Hayden & Platt, 2007, 2009; S. R. Heilbronner & Hayden, 2013; O'Neill & Schultz, 2010). In contrast, a recent study found mild risk aversion in rhesus monkeys (Yamada, et al., 2013). Compared to the previous studies, this one had higher reward amounts, non-rewarded trials, multiple comparisons with unequal expected utilities, and visual cues (pie charts) to represent the available reward amounts. It thus remains unclear what factors lead to risky versus safe preferences in rhesus monkeys. In Experiments 1–6, I examined the effect of several factors while keeping the expected utilities of the choice options equal (as they were in the original studies that found preferences for the risky options). More specifically, I examined the following: (1) to change reinforcer type as well as to examine the effects of reward quality, I used food rather than fluid reward; (2) to increase the stakes with respect to the variation in the risky options, I heightened the risk level beyond those used in previous studies; (3) to increase the cost of risky choices, I (a) used no reward outcomes (omission of reward), and (b) reduced the likelihood of receiving the large reward outcome associated with the risky option to below 50%; and (4) to increase the significance of individual choices, I

used (a) a maximum of 32 daily free choice trials per monkey (as opposed to hundreds of free choice trials), and (b) a longer intertrial interval (15 s). Despite these modifications, both monkeys showed a consistent preference for the risky option except when the expected value for the safe option was greater than for the risky option, and in this case, fourfold greater. Future testing can systematically modify the mean utility between the safe and risky options to determine the indifference (subjective equivalence) point of such preferences.

Because fluid rewards were previously used (Hayden, et al., 2008; Hayden & Platt, 2007; McCoy & Platt, 2005; O'Neill & Schultz, 2010), these results show that the monkeys' risk preference extends to food-based rewards as well. In addition, Hayden and Platt (Hayden & Platt, 2007) found that the rhesus monkeys' preference reached neutrality as the intertrial interval increased to 90 s in their gambling task. In this study, I used an intertrial interval of 15 s. However, because the risky option used reward omission outcomes instead of small-reward outcomes, the monkeys experienced a further delay similar to an additional intertrial interval. Because the large reward outcome had a lower likelihood (e.g., 12.5% in Experiment 5 compared to 50% in the study of Hayden and Platt), the monkeys easily experienced many inter-reward intervals >90 s; and yet they still maintained a risky preference.

One possible account for the risk preference of the monkeys in this study was that the initial preference for the risky option in Experiment 1 drew their preference toward the risky option for the subsequent Experiments 2–6. It has been shown that choices themselves can influence subsequent preferences (D. Kahneman, 2011; Klos, 2005; Montgomery & Adelbratt, 1982; Rabin & Vayanos, 2010; Sundali & Croson, 2006),

especially when subjects must learn the outcome likelihoods (i.e., probabilities) via experience over trials. Although interactions among experiments could influence choices, we think that this factor was minimal in this study. First, I reversed the left–right configurations within each experiment while holding the visual stimulus locations constant to mitigate against simple stimulus–response associations driving choice. Thus the monkeys’ preferences needed to be based on the actual reinforcement contingency in each experiment independent of visual stimuli and location. Second, to help demarcate the change in the reinforcement contingencies across the experiments (and configurations within each experiment), I used a forced and free choice block structure in which the monkeys received four consecutive forced choice trials with each option, followed by eight free choice trials. The forced choice trials help provide clearer feedback with respect to the current reinforcement contingencies. Third, if the monkeys were continuing to use the same general likelihoods across experiments, the monkeys should not have required so many trials to develop a significant preference in Experiments 2–6: an average of 35 (Monkey P) and 55 (Monkey T) per configuration, 69 (Monkey P) and 109 (Monkey T) per experiment (see **Table 7**); and further, they should have exhibited a clear decrease in the number of trials to attain a significant preference across Experiments 1–6, which they did not. Fourth, Experiment 4, which replicated Experiment 2, provided a test for the effect of experience. The acquisition rates did not differ for either monkey. Finally, the overall faster acquisition rate of both monkeys in Experiment 5 (which had the largest payoff outcome of the more- preferred food item), as well as the reversal of preference to the safe option in Experiment 7, showed that the monkeys were influenced by the actual reward contingencies of each experiment. Thus,

the continued risk preferences appeared to reflect a preference for the given option under each of the experimental contingencies.

Therefore, the risk preference generalizes across multiple contexts. Yet it remains unclear why a strong risk preference was found here, whereas a mildly risk averse preference was found in the Yamada et al. study (Yamada, et al., 2013). There appear to be three leading possibilities. One, because I baited the risky trials (see **1Bi.1.2 MATERIALS AND TESTING PROCEDURES**), ensuring that the reward would occur within a certain number of trials, it is possible that the dependence of risky choice outcomes affected decision-making by reducing the negative impact of the non-rewarded trials (although no direct impact on behavioral strategies was observed). Second, experience with fixed probabilities and multiple risky values might help to train the monkeys to respond more perceptively to both probability and value and thus according to expected value. Finally, the use of pie charts to represent the available reward amounts during decision-making might help to clarify and simplify the reward estimation. It certainly should help to minimize memory effects. Further work will be necessary to determine the contributing factors.

Thus, taken together, contextual factors clearly influence decision-making preferences (S. R. Heilbronner & Hayden, 2013). Nonetheless, inherent preferences might still exist, leading to species and individual differences. Such potential biases need to be examined under minimal experience and with multiple measures including acquisition rates. It also will be critical to compare groups in testing paradigms as identical as possible. To that end, people were tested in a comparable paradigm to that used with rhesus monkeys that obtained risk-seeking in the monkeys (Hayden & Platt,

2009). The human subjects appeared to be less loss averse than in instructed tasks, with the majority of people being risk-neutral (i.e., no significant preference found between safe and risky options). In fact, three general populations could be identified: risk-averse, risk-neutral and risk-seeking. In addition, in a similar paradigm, differences were found between human children, adolescents, and young adults, with children showing a preference for risky over safe options, adolescents showing a mild risk preference, and young adults generally showing a preference for the safe option (Paulsen, Carter, Platt, Huettel, & Brannon, 2011; Paulsen, Platt, Huettel, & Brannon, 2011, 2012). Thus, human risk preferences cannot be simply explained by a general context difference such as instruction- vs. experience-based choices.

It is of interest that rhesus monkey risk preferences may resemble those of some human children, adolescents, and adults, at least to some degree (Hayden & Platt, 2009; Paulsen, Carter, et al., 2011; Paulsen, Platt, et al., 2011; Paulsen, et al., 2012). It is worth exploring whether this finding results from similar relative differences in the balance of control of brain structures involved in higher-level integration of value across trials versus those driving impulsive responses (Delgado, Gillis, & Phelps, 2008; Hare, Camerer, & Rangel, 2009; Hassin, Ochsner, & Trope, 2010; Kober, et al., 2010; McClure, Laibson, Loewenstein, & Cohen, 2004; Volkow, 2010; Volkow, et al., 2010). A preference for risky options in some non-human animals, human children, adolescents, and adults could reflect relatively weak executive control.

More specifically, general risk preferences such as what we observed in Experiments 1–6 could be due to value estimation that is non-linear. That is, the actual value added per reward item might increase with an increasing number of items. In other

words, two M&Ms might have more than twice the value of one M&M. Some evidence suggests that rhesus monkeys are significantly more sensitive to changes in high-reward outcomes compared to low-reward outcomes (Hayden, et al., 2008). Our results agree. The fastest acquisition rate occurred in Experiment 5, in which the monkeys could obtain the largest amount of the preferred food item in any experiment – eight mini M&Ms – even though it was only received 12.5% of the time. Similarly, risk preferences could reflect insensitivity to the lower-valued risky outcome, especially when the value of the outcome is lower than the potential “jackpot,” but nonetheless positive or 0, as opposed to a negative value associated with an aversive outcome. Risk preferences could also reflect a relative insensitivity to the likelihoods, and in consequence, to the risk being taken (D. Kahneman, 2011; D. Kahneman & Tversky, 1979). Finally, risky choice behavior could also reflect a preference for outcome stochasticity (Hayden, et al., 2008). Future work will be needed to tease apart the potential influences on risky behavior such as a fixation on the highest outcomes, a relative insensitivity to non-rewarded or low cost outcomes, a relative neglect of likelihood, a preference for outcome stochasticity, and other factors.

As opposed to risk preferences reflecting poor behavioral control, another possibility is that a preference for risk evolved as an adaptation for exploratory behavior in foraging or social behavior. Risk preference and impulsiveness in rhesus monkeys could reflect ancestral conditions that selected for such behavior, such as an omnivorous diet that might provide a selective advantage for the seeking of new outcomes of foraging choices or the heightened conspecific competition in a rigidly hierarchical social system (T. A. Evans & Beran, 2007; J. D. Kralik & Sampson, 2012; Passingham & Wise, 2012;

Thierry, et al., 2004; Wise, 2008).

At the same time, there is certainly diversity and individual variability in non-human primates and people, and selection forces sculpting decision-making strategies may have led to multiple response “niches,” with relatively bolder and more prudent individuals both finding success. Indeed, even our two monkeys exhibited differences in their choice behavior: e.g., the effect of reward quality, and the use of response strategies in Experiment 7. Nonetheless, general species differences are also possible. Because our two subjects revealed the same general preferences in all seven experiments and because our results match other studies showing risk preferences in rhesus monkeys (Hayden, et al., 2008; McCoy & Platt, 2005), our findings might represent rhesus monkey preferences in general, at least under our task parameters. Future work with more subjects will be necessary to determine species-level risky preferences and characterize the circumstances they may manifest. A two-subject study does however reveal a capacity within the species among at least a subpopulation of individuals, especially when corroborated by other published findings.

Finally, we found some evidence for the influence of reward quality on preference: risk preferences decreased with the decrease in reward quality in Experiment 6 versus 4 (significantly so for monkey P). A heightened general reward level in the environment can increase the motivational level of the individual, leading to increased response vigor (Niv, Daw, Joel, & Dayan, 2007). This finding corroborates those that have found increased risk-seeking with increased motivation, such as hunger, but contrasts with other findings (Caraco, et al., 1980; e Abreu & Kacelnik, 1999; Kacelnik & Bateson, 1996, 1997; Yamada, et al., 2013). To be sure, more work is necessary to

clarify the effects of related variables such as motivation, “wealth,” and general environmental conditions on risk behavior.

1Bi.3.2 Safety preference: Experiment 7

Despite the risk preference shown in Experiments 1–6 and in published work (Hayden, et al., 2008; McCoy & Platt, 2005; O'Neill & Schultz, 2010), the results in Experiment 7 showed that the risk preference could be overcome, and that it was not simply due to factors such as the neglect of likelihood or the insensitivity to trials with no reward. Ultimately, both monkeys demonstrated the capacity to integrate likelihood and value across trials to obtain an expected value for the risky option, which was lower than the safe option in Experiment 7. The incentive required to induce monkeys to shift from risky to safe options needs further study, especially since our two monkeys appeared to use different strategies. Monkey T used a win-stay, lose-shift strategy, but Monkey P did not. In any case, Experiment 7 shows that rhesus monkeys have the capacity to override any impulsive tendency to select the option with the highest possible outcome under our experimental conditions, as was found in the Yamada et al. study (Yamada, et al., 2013).

1Bi.3.3 CONCLUSION

Affective decision-making is a fundamental cognitive process, with multiple phenotypic variations in people (Heino, vanderMolen, & Wilde, 1996; Daniel Kahneman & Tversky, 2000; Paulsen, Platt, et al., 2011; Schuck-Paim, Pompilio, & Kacelnik, 2004; Soane & Chmiel, 2005; E. U. Weber, Blais, & Betz, 2002; Weller & Tikir, 2011), and with significant personal, societal, and clinical ramifications when risk-seeking becomes excessive (Admon, et al., 2012; Davis, Levitan, Muglia, Bewell, & Kennedy, 2004; Fishbein, et al., 2005; Kreek, Nielsen, Butelman, & LaForge, 2005; Rasmussen & Eisen,

1992). An evolutionary and comparative analysis should help delineate the nature of its underlying component processes and the conditions under which different phenotypes are manifested.

In the present study, we found that rhesus monkeys prefer risky to safe options in many circumstances, at least when the expected value of the safe and risky options are equivalent. How this risky preference interacts with utility maximization is yet to be determined. Rhesus monkeys are descendants of the last common ancestor (LCA) of Old World monkeys, apes, and humans. Assuming that their capabilities resemble the LCA, at least more closely than modern humans do, this catarrhine species' abilities may reflect the evolution of cognition in our lineage at a time when both a lower-level impulsive process and a higher-level, more integrative process existed, with perhaps a bias toward the former. Given that a general dual-process architecture is hypothesized to underlie human cognition (Antoine Bechara, 2005; Dolan & Dayan, 2013; J. S. B. Evans & Stanovich, 2013; D. Kahneman, 2011), behavioral and neurophysiological studies with rhesus monkeys should help characterize the dynamics of the underlying cognitive processes (see **Chapters 2 and 3**), as well as provide insight into the evolutionary history of decision-making in a risky and uncertain world.

1Bii. Fluid-Based Gambling Task

To further delineate the key factors influencing affective decision-making during decision-making under risk, we developed and conducted a novel fluid-based gambling task similar to my food-based gambling task (**Chapter 1Bi.**), and conducted it with two rhesus monkeys I did not use in the food-based gambling task. In this task, the monkeys were trained to choose between a safe and a risky option for a number of squirts of fluid as reward. To assess the ability of rhesus monkeys to integrate value and likelihood in assessing expected value, we conducted two conditions where the safe option had a different overall expected value/utility with the risky one. Both monkeys were able to integrate value and likelihood to guide their choices and preferred the option with the higher expected value. To assess the influence of the risk alone (variance) in affecting preference, I conducted another condition where the safe and risky option had equal expected value. While one monkey had a significant preference for the risky option in condition 3, the other monkey did not show a significant preference for either the risky or safe option.

1Bii.1. MATERIALS AND METHODS

1Bii.1.1 Subjects

Two male rhesus monkeys were available for this study, denoted as Monkey C and H, ages 12 and 11, respectively. Monkey C and H were not used in the food-based gambling task. Animal care and use complied with all current laws, regulations, policies, and guidelines of the United States, the United States Department of Agriculture (USDA), the

Public Health Service (PHS), and all procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of Dartmouth College. The Center for Comparative Medicine and Research (CCMR) at Dartmouth maintains a full-time animal care and veterinary staff that monitors the monkeys' daily health and well-being.

The monkeys were housed in a homeroom with automatically regulated temperature, ventilation, humidity, and lighting. The monkeys were intermittently housed together and individually: at times when they engaged in confrontations, which is normal periodic behavior in young rhesus macaque males of similar size and temperament (Thierry, et al., 2004), the two monkeys were separated and individually housed for their safety. When pair-housed, they had direct physical contact with each other, and when individually housed, through a mesh grading divider between their cages. They also had direct visual contact with the other monkeys in the colony, as well as the animal care staff and experimenters. In addition, environmental enrichment included two or more enrichment items in their home cages at all times, daily playing of radio or videos in the room (the latter via a monitor mounted in view of all individuals), and regular access to a larger enrichment cage (172.7 cm × 96.5 cm × 182.9 cm) in an adjacent room. The monkeys had *ad libitum* access to food but were water-restricted during the study to ensure high motivation.

1Bii.1.2 Materials and testing procedures

Upon entering the laboratory, the monkeys sat in a restraining chair with his head fixed and his eye position tracked using the Arrington viewpoint eye tracking system (Arrington Research, Scottsdale, AZ). The monkeys sat in front of a computer monitor

(40.5 cm from Monkey C, 26 cm from Monkey H) to perform the fluid-based gambling task.

The gambling paradigm used in this study (see **Figure 8**) is an adaptation of the visual gambling paradigm used by McCoy and Platt (McCoy & Platt, 2005). During a forced choice trial (see **Figure 8** top row), each trial began with a fixation point, a small white circle ($0.9/2.6^\circ$ visual angle for monkey C/H, respectively), that appears at the center of the monitor screen along with one of the two saccade targets (4.8° by $4.8^\circ/7.5^\circ$ by 7.5° squares for monkey C/H, respectively). During a free choice trial (see **Figure 8** bottom row), the trial starts with a fixation point at the center of the monitor screen along with two saccade targets. Once the monkey fixates his eye position on the circle, it turns gray in color as feedback. To proceed, the monkey must maintain fixation of the gray circle for 500 milliseconds (ms). If the monkey's gaze exits fixation, the fixation circle turns back white and the task does not proceed. During successful fixation, the gray circle disappears, serving as a "go" cue for the monkey to initiate a saccadic eye movement to the target (for a forced trial) or either one of the two colored targets (for a free choice trial). Once the monkey reaches a target, the unchosen target disappears, and the monkey must maintain target fixation of the chosen target for 25ms. Once again, the task does not proceed until the monkey maintains fixation of his first chosen target for 25 ms. After a successful 25 ms target fixation, there was a 500 ms pre-reward delay before the monkey received a number of squirts of juice (1, 4, or 7 squirts) based on the reward condition contingencies (see below and **Tables 8 and 9**). All trials were equalized in length after reward delivery so that a smaller/larger reward did not result in the next trial starting earlier/later, respectively. The intertrial interval was 1100 ms.

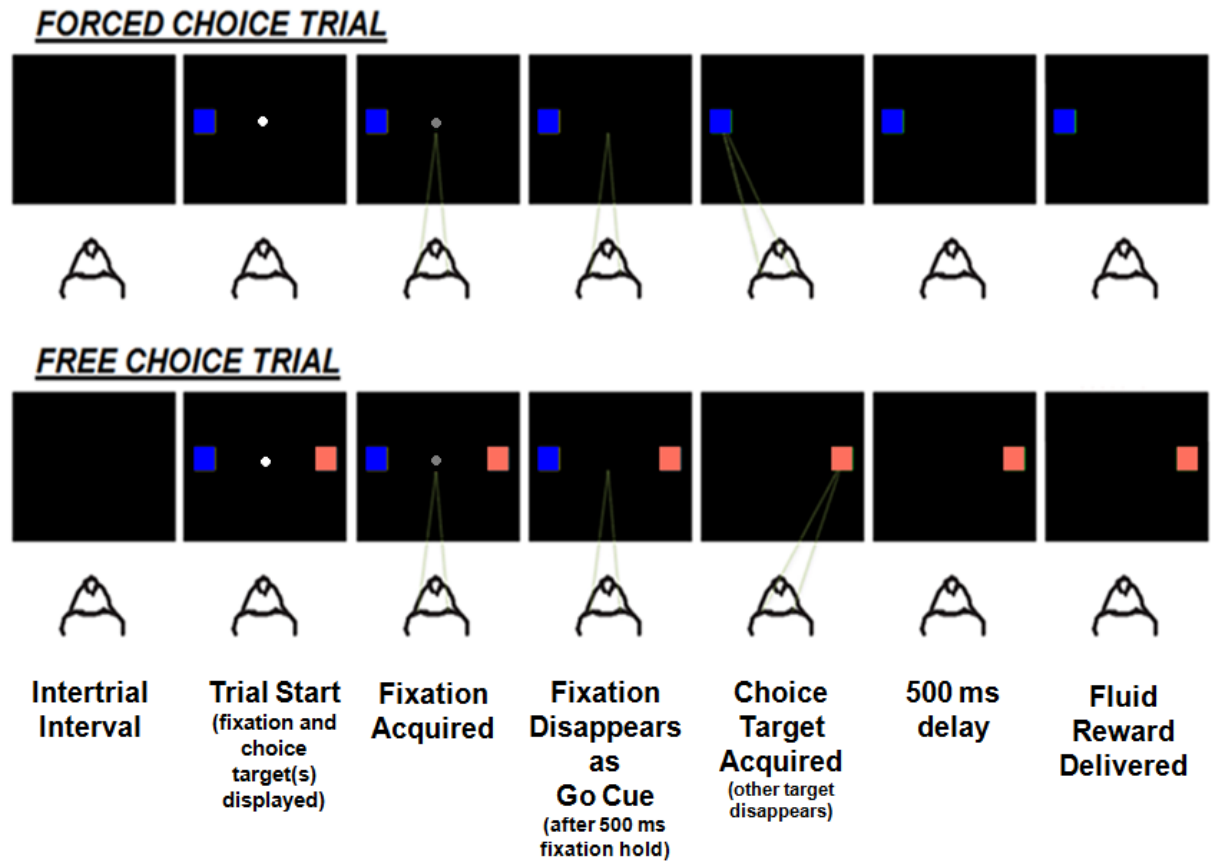


Figure 8. Fluid-based gambling task (1Bii): Task and event schematic (from monkey's point of view of computer monitor) in one forced choice trial to the left target (top) and a free choice trial (bottom). The fluid-based gambling task was run in blocks of 16 trials: eight forced choice trials—two forced choice trials from one of the two targets (randomly determined at the beginning of the block) followed by two forced choices from the other target, followed again by two forced choices from the first experienced target, and finally two forced trials from the other target again—and eight free choice trials.

There were three conditions in this task. In our first condition, the risky option had a higher expected value relative to the safe option; whereas, in the second condition the risky option had a lower expected value. In our third condition, the risky and the safe option both had equal expected value (see **Table 8**). Each free choice trial consisted of a safe option and a risky option stimulus (see **Table 9**). The safe option always delivered 4 squirts of juice, while the risky option delivered either 1 squirt or 7 squirts of juice depending on the condition. In condition 1, the risky option delivered 1 squirt or 7 squirts of juice 12.5% and 87.5% of the time, respectively. In condition 2, the risky option delivered either 1 squirt or 7 squirts of juice 87.5% and 12.5% of the time, respectively (the opposite of condition 1). Finally in condition 3, the risky option delivered 7 squirts or 1 squirt of juice with equal likelihood, 50% and 50%. The likelihoods of the risky choice outcomes were not baited and thus the risky outcomes were independent across trials.

The fluid-based gambling task was run in blocks of 16 trials: eight forced choice trials—two forced choice trials from one of the two targets (randomly determined at the beginning of the block) followed by two forced choices from the other target, followed by two forced choices from the first target, and finally two forced trials from the other target—and eight free choice trials. We used the forced choice trials conducted at the beginning of every block as the means by which the monkeys learned the contingencies of any given condition. To rule out the potential influence of location biases and to further confirm the monkeys' true preferences, each condition was conducted in two configurations in which the safe and risky targets switched positions on the monitor screen. Per daily session, the monkeys were given a randomly determined configuration

of either condition 1 or condition 2. To ensure that the monkeys were contacting the contingencies and properly assessing value and likelihood, the monkeys were required to show a significant risk preference for condition 1 or a significant safe preference for condition 2 for his last 12 free choice trials ($\geq 10/12$, $p < 0.05$, two-tailed binomial test). After the monkey reached the criterion, the configuration of the risky and safe option switched within the same condition. The monkeys were then required to show the same preference for the other side. This procedure was repeated with the other condition (1 or 2) under both configurations with the same criteria. After the monkey successfully completed condition 1 and 2, the monkey experienced condition 3. Because the utility of the safe and risky option were the same in this condition, there was no preference criterion. The monkey simply experienced 32 total trials in each configuration (64 free and forced choices combined).

Condition	Safe Option	Risky Option		
		Contingencies	Variance	Utility/Expected Value
1	4 squirts (100%)	7 squirts (87.5%) & 1 squirt (12.5%)	3.92	6.25 squirts
2		7 squirts(12.5%) & 1 squirt (87.5%)	3.92	1.75 squirts
3		7 squirts (50%) & 1 squirt (50%)	9	4 squirts

Table 8. Fluid-based gambling task (1Bii): Contingencies for the safe and risky option for all 3 conditions. Each condition consisted of a choice between a safe and a risky option. The safe option was the same across all three conditions.

	Monkey C	Monkey H
Safe Option		
Condition 1 Risky Option		
Condition 2 Risky Option		
Condition 3 Risky Option		

Table 9. Fluid-based gambling task (1Bii): Square target stimuli used for the safe and risky choice for each monkey across all 3 conditions.

1Bii.2.1 Choice Preferences

During free choice trials, both monkeys had the same general preference for conditions 1 and 2. In condition 1, both monkeys showed a significant preference for the risky option with the higher expected value (one sample t-test against 50%, two-tailed: monkey C: 82%, $t(1731)=35.16$, $p<0.001$; monkey H: 69%, $t(269)=6.86$, $p<0.001$). In condition 2, both monkeys showed a significant preference for the safe option with the higher expected value (one sample t-test against 50%, two-tailed: monkey C: 89%, $t(1430)=47.86$, $p<0.001$; monkey H: 74%, $t(243)=7.89$, $p<0.001$). In condition 3, monkey C showed a risk preference (one sample t-test against 50%, two-tailed: 63%, $t(1206)=9.517$, $p<0.001$) while monkey H did not (one sample t-test against 50%, two-tailed: 48%, $t(204)=0.49$, $p=0.62$). Nonetheless, both monkeys showed the same trend: the highest risk-preference in condition 1 and the highest safe-preference in condition 2.

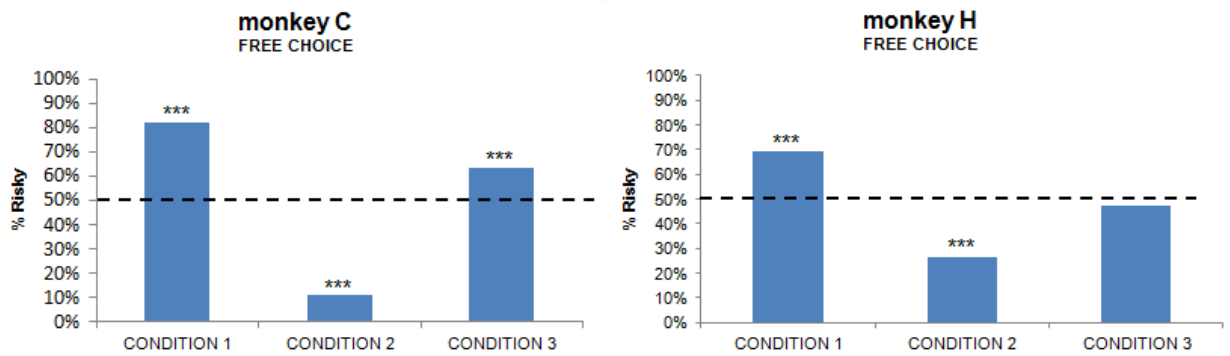


Figure 9: Fluid-based gambling task (1Bii): Percentage of free choices where each monkey chose the risky option for all three conditions. The dashed line at 50% indicates no preference between the risky and safe option. Asterisks: significant preference (two-tailed one sample t-test, *** $p < 0.001$).

1Bii.2.2 Response Times

The response times were measured as the latency between successful fixation (go-cue) to when the choice target was acquired (see **Figure 8**). Linear correlations were measured to determine the influence of an option's expected value and variance (risk) with response time. Across sessions, the monkeys showed a negative correlation of response time with expected value, such that the monkeys were fastest to saccade toward targets of high expected value and slowest for low expected value targets (both monkeys, correlation coefficient: -0.17, $p < 0.0001$; monkey C, correlation coefficient -0.19, $p < 0.0001$; monkey H, correlation coefficient: -0.08, $p = 0.29$). There was no significant correlation between variance and response times (both monkeys, correlation coefficient: 0.00093, $p = 0.92$; monkey C, correlation coefficient -0.000059, $p = 0.99$; monkey H, correlation coefficient: 0.0098, $p = 0.42$).

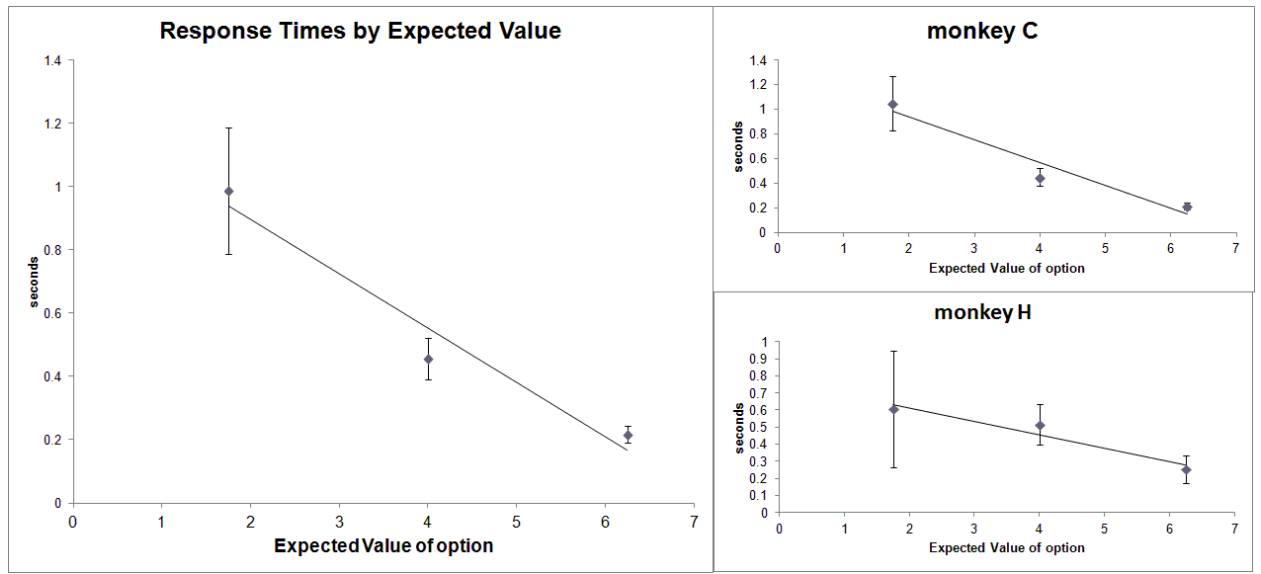


Figure 10: Fluid-based gambling task (1Bii): Average response times of the monkeys

(left: monkeys combined; top right: monkey C only; bottom right: monkey H only) across sessions as a function of the average expected value of the options (1.75 (risky option condition 2); 4 (safe option + risky option condition 3); 6.25 (risky option condition 1)).

There was an overall negative correlation of response time with expected value significant with both monkeys combined ($p < 0.0001$) and with monkey C alone ($p < 0.0001$).

1Bii.3 CONCLUSION & DISCUSSION

Previous studies using fluid reward to study risky decision-making have used juice volume to alter the value of rewards and 50/50 likelihoods for low and high outcomes from the risky option (Hayden, et al., 2008; Hayden & Platt, 2007, 2009; McCoy & Platt, 2005; O'Neill & Schultz, 2010). Our study used discrete numbers of juice squirts to increase the salience of the value of outcomes and outcome likelihoods greater and lower than 50% so that likelihood played a larger role in determining expected value. Our findings show that the monkeys were able to integrate value and likelihood to choose the option with higher expected value (conditions 1 and 2). For condition 3, one monkey showed a preference for the risky option while the other did not show any significant preference. Behaviorally, the monkeys showed a significant negative correlation of response time with the expected value of the option. Therefore, the monkeys were fastest at acquiring targets with large expected values and slowest at acquiring targets with low expected value (at least for monkey C). Response times did not significantly vary with the variance (our measure of risk) of the options.

There are some important differences between the food-based and the fluid-based gambling tasks. The monkeys in the food-based gambling task were neither food nor water restricted while the monkeys in the current fluid-based gambling task were water restricted. Another important difference was the intertrial interval used. The food-based gambling task used an intertrial interval of ~8 seconds and this task used an intertrial interval of 1.1 seconds. The literature suggests that these changes would increase risky preferences in the fluid-based gambling task due to a decrease in “wealth” in the leaner, water-restricted environment (Kacelnik & Bateson, 1996, 1997; Yamada, et al., 2013)

and a lower temporal discounting of the large outcomes from the lower intertrial interval (Hayden & Platt, 2007). Interestingly, however, monkey H did not show a risky preference as predicted.

The monkeys ultimately seem to be driven by expected value of the options; however, when the expected value between options was the same, 3 of the 4 monkeys across the two experiments (monkey C, P, T) exhibited a preference for the risky option. One possible reason for the lack of a risk preference by monkey H in the fluid-based gambling task is the addition of a condition where the risky option had a higher expected value than the safe option (condition 1 in this case). By adding a condition where it was advantageous to risk, it may have decreased the monkeys' propensity to risk in other conditions. To put it more simply, the monkeys may have quelled their desire for the large outcome of the risky option if there is knowledge of other conditions where it was more opportune to risk. The literature seems to suggest this is the case, as rhesus monkey studies where there was no such risky advantageous condition found a risk preference (Hayden, et al., 2008; McCoy & Platt, 2005; O'Neill & Schultz, 2010), whereas studies that found no risk preference did have risky advantageous conditions (Yamada, et al., 2013). Before the addition of condition 1, monkey C was run in sessions of just conditions 2 and 3. Comparison of risky behavior in condition 3 before and after the addition of condition 1 show a significant reduction in risky choice from 72 to 63% (one sample t-test against 50%, two-tailed: $t(1789)=3.767$, $p=0.0002$), respectively. Although contextual influences on risk-seeking have been mentioned in the literature (S. R. Heilbronner & Hayden, 2013), none have discussed this particular contextual effect

influencing risky decision-making. It will be interesting in the future to study this effect as a potentially critical modulator of risky preferences.

Chapter 2.

Neural Processing of Risky Decision-Making by the Anterior Insula Cortex

2.1 ANTERIOR INSULA

The anterior insula is a bilateral subcortical structure lying behind the lateral fissures of the brain in the posterior part of the frontal cortex. It is an agranular area that lacks a conspicuous layer 4 of cell bodies. This structure has been thought of as an affective brain area involved in awareness and affective feeling states (A. Bechara & Damasio, 2005; Damasio, et al., 2000; Singer, Critchley, & Preuschoff, 2009), integrating cognition with emotion (Berntson, et al., 2011; Craig, 2009; Critchley, 2009), and appears to have a powerful role in how we evaluate options. How we evaluate options ultimately affects our choices under risky conditions.

The anterior insula is situated in an area that promotes evaluative processing. The anterior insula cortex receives dopaminergic input from the ventral tegmental area and is comprised of well-documented dopaminergic receptors (Gaspar, Bloch, & Le Moine, 1995; Richfield, Young, & Penney, 1989). These mesolimbic dopaminergic neurons in the midbrain of the rhesus monkey have been found to encode relative reward value (Tobler, Fiorillo, & Schultz, 2005) and scale with reward likelihood (Fiorillo, Tobler, & Schultz, 2003). Anatomical studies with rhesus macaques revealed efferent projections from the anterior insula to the ventral striatum (Chikama, McFarland, Amaral, & Haber, 1997; Fudge, Breitbart, Danish, & Pannoni, 2005) where subregions (of the ventral striatum) have been found to encode reward value and likelihood (Yacubian, et al., 2007). Because dopamine and mesolimbic dopamine projections to the ventral striatum have been found to play a central part in reward processing (Naqvi & Bechara, 2009), the anterior insula is positioned to have a critical influence on decision-making and reward processing.

The anterior insula has been found to hold representations of positive value. In the brain damage literature, Berntson and colleagues (Berntson, et al., 2011) found that people with anterior insula damage had attenuated arousal and valence ratings for positive picture stimuli compared to controls. Furthermore, damage to the insula reduced the positivity of previously addictive substances such as cigarettes and cocaine (Contreras, Ceric, & Torrealba, 2007; Naqvi, Rudrauf, Damasio, & Bechara, 2007). With respect to decision-making paradigms, lesions have been found to diminish an individual's sensitivity to the overall expected value of outcomes in the gain domain (Weller, Levin, Shiv, & Bechara, 2009) such that these individuals took significantly less risks (even when it was advantageous to do so) compared to controls. In rats, inactivation of the anterior insula reduced preferences for risky options when both the safe and risky option had equal expected value (Ishii, Ohara, Tobler, Tsutsui, & Iijima, 2012). In human neuroimaging, anterior insula activity has also been found to precede a risky choice for gambles between options of equal expected value (Xue, Lu, Levin, & Bechara, 2010).

Conversely, the anterior insula has also been found to hold and intensify representations of negative value. Berntson and colleagues (Berntson, et al., 2011) found that people with anterior insula damage had not only attenuated arousal and valence ratings for positive picture stimuli, but to negative ones as well. Lesions of the anterior insula have also been found to also diminish an individual's sensitivity to the likelihood of losses when making bets in the loss domain (Clark, et al., 2008; Weller, et al., 2009). These individuals ended risked more than controls and did not adjust their decisions closely with expected value. The anterior insula also appears to signal undesirable

representations in the environment, such as anticipated risk, disgust, and drug withdrawal (Calder, Keane, Manes, Antoun, & Young, 2000; Naqvi & Bechara, 2009; Preuschoff, Quartz, & Bossaerts, 2008). Microstimulation of this brain region (to increase neural activity), produced disgust reactions in rhesus monkeys toward previously positive appetitive food items (Caruana, Jezzini, Sbriscia-Fioretti, Rizzolatti, & Gallese, 2011). In neuroimaging, anterior insula activity has also been found to precede safe choice behavior during investment decisions between safer bonds and riskier stocks (Knutson, Rick, Wimmer, Prelec, & Loewenstein, 2007; Kuhn & Knutson, 2005).

In general, the anterior insula appears to have a central role in processing subjective utility and influencing affective decision-making. However, some findings appear to point to a role in reward processing which may enhance risky decision-making, while others in the control of reward processing centers which can promote safer decision-making. Due to the conflicting findings in the literature, more evidence is needed to determine the exact role of neurons in the anterior insula in decision-making under risk. Therefore, in the current study, we use the fluid-based gambling task to examine how neurons in the anterior insula correlate with variables such as expected value, relative value, and risk during affective decision-making.

2.2 NEUROPHYSIOLOGY METHODS

A chamber was mounted over the left hemisphere of the brain that allowed subcortical access to the anterior insula (and premotor cortex) of monkeys C and H. The placement of the chambers was determined using stereotaxic coordinates as determined by a published rhesus monkey atlas (Paxinos, Huang, & Toga, 2000). Single-unit

extracellular recording was performed using glass (Alpha Omega, Inc., Alpharetta, GA) or epoxylite-coated (FHC, Inc., Bowdoin, ME) tungsten single microelectrodes. A NAN drive system (NAN Instruments, Inc., Nazareth, Israel) and a custom-made grid (spaced 1 mm apart) was used to position the microelectrodes in the XY plane of the chamber opening. The microelectrodes were held in electrode towers attached to a base sitting on top of the monkey's chamber. Each microelectrode was held within an electrode tower by an electronically-moveable electrode holder and a manually-moveable guiding tube cannula. To ensure microelectrode stability, the guiding tube cannula was lowered and threaded through the custom-made grid until the cannula contacted the monkey's dura. Using a computer program, the microelectrode was then lowered precisely via a miniature motor that was connected to the tower.

Signals from the electrodes were amplified and filtered and then sent directly to the multi-channel acquisition system (Plexon, Inc., Dallas, GA). Microelectrode placement depth was determined using a combination of the rhesus monkey atlas (Paxinos, et al., 2000), each monkey's structural MRI, the electrode height on the tower, and audio-visual evidence from the multi-channel acquisition system. To ensure neurons were well isolated, stable, and well separated from noise, neuronal waveforms were sorted off-line (Offline Sorter, Plexon, Inc., Dallas, GA) using template matching, principal component clustering analysis, and by analyzing signal-to-noise ratios and interspike interval histograms. Spike and event timestamps were sent to Matlab for analysis.

The materials and testing procedures were the same as Chapter 1Bii (see **1Bii.1.2 Materials and testing procedures**) as well as the task used (fluid-based gambling task).

To ensure that the neurons were responding to the task contingency, only the last 20 trials of each configuration of each condition were used for analysis. In condition 1, this would be the last 20 trials after which the monkey showed a significant risk preference in his last 12 free choice trials ($\geq 10/12$, $p < 0.05$, two-tailed binomial test) for that configuration, and again for the other configuration of condition 1. In condition 2, this would be the last 20 trials after which the monkey showed a significant safe preference for that configuration across his last 12 free choice trials ($\geq 10/12$, $p < 0.05$, two-tailed binomial test), and again for the other configuration of condition 2. In condition 3, this would be the last 20 trials in the 32 trials run in each configuration. For a neuron stable across all 3 conditions, this would be a total 120 trials. Although monkey C and H showed a behavioral difference in condition 3 whereby monkey C was significantly risky and monkey H was risk-neutral (see **1Bii.2.1** and **Figure 9**), both monkeys showed the same individual trend, namely, the highest risk-preference in condition 1 and the highest safe preference in condition 2. This same trend showed that the monkeys were contacting the contingencies relative to each other in an equivalent manner. Thus, we pooled neurons from both monkeys in the analysis. Overall, we sampled electrophysiological activity from 52 single anterior insula neurons (46 & 6 in monkey C & H, respectively) while the monkeys performed the fluid-gambling task (see **Chapter 1bii** for details). Of these 52 neurons, 30 were stable across all 3 conditions. All 52 were analyzed when appropriate.

Analysis was conducted to determine whether firing rates were related to the following variables (A-D):

A. Expected value (EV) – the mean expected value/utility of the options by objective calculations. This value was 2.75 for the risky option of condition 2, 4 for the safe option and the risky option in condition 3, and 6.25 for the risky option in condition 1.

B. Variance – the square of the standard deviation of choice outcomes. This value was 0 for the safe option, 3.92 for the risky option in conditions 1 and 2, and 9 for the risky option in condition 3.

C. Relative safe – The relative value of the safe option in comparison with the risky one. That is, even though the safe option always had an expected value of 4, its relative value compared to the particular risky option differed across conditions: from lowest to highest: condition 1 → condition 3 → condition 2.

D. Location – To test for coding of the location of the two choice option targets.

In addition, variables A-D were analyzed in five time periods or windows across the trial (see **Figure 8** for reference):

1. **FIXATION** – Defined as: ([go cue – 500 ms] to [go cue]). During this period valuation and possibly decision-making occurs.

2. **CHOICE** -- ([go cue] to [choice target acquired]). During this period the actual choice is made (action-selection).

3. PRE-REWARD --

([1st drop of fluid reward – 500 ms] to [1st drop of fluid reward]). This is the period just prior to receiving the first reward squirt and serves as a reward anticipation period.

4. REWARD 1 -- ([1st drop of fluid reward] to [1st drop of fluid reward + 500 ms]). The first reward squirt is received in this period.

5. REWARD 2 --

([2nd (potential) drop of fluid reward] to [2nd (potential) drop of fluid reward + 500 ms]).

This is a potential feedback period when the risky outcome is resolved (i.e., receiving nothing signals the low outcome of 1 squirt, while receiving a squirt signals the jackpot of 7 squirts).

Variables A-C above were analyzed using linear regression; variable D was analyzed using a 2-sample t-test. Once the percentage of significant neurons was determined for each variable in each time period, I used a one-tailed, one-sample t-test to determine if the obtained percentage was significantly higher than expected by chance (i.e., whether greater than 5%). I then classified the significant correlations for each variable in each time period as either positively or negatively correlated, and used a one-tailed, one-sample t-test to determine if these percentages were significantly higher than expected by chance (i.e., whether greater than 2.5%).

2.3 RESULTS

The results for the main variable correlations (**Figure 11**) show that there were

significant correlations of all variables in the task in at least one of the analysis windows ($p < 0.05$, one-tailed, t-test of percentage against chance (5%)). With respect to expected value, there was a significant percentage of neurons that coded this variable in the choice, pre-reward, and reward analysis windows (choice: $t(47)=2.04$, $p=0.024$; pre-reward: $t(47)=3.78$, $p=0.0002$; reward: $t(47)=2.50$, $p=0.0081$; reward 2: $t(47)=3.78$, $p=0.0002$). For variance, there was a significant percentage of neurons that were correlated with this variable in the pre-reward and reward windows (pre-reward: $t(42)=2.40$, $p=0.011$; reward 1 & 2: $t(42)=1.91$, $p=0.031$). For relative value of the safe option, there was a significant correlation in all analysis windows (fixation: $t(42)=2.16$, $p=0.019$; choice & pre-reward: $t(42)=2.40$, $p=0.011$; reward 1: $t(42)=2.63$, $p=0.006$; reward 2: $t(42)=2.86$, $p=0.0033$). Lastly, there was significant location selectivity in the choice analysis window ($t(51)=1.7$, $p=0.047$).

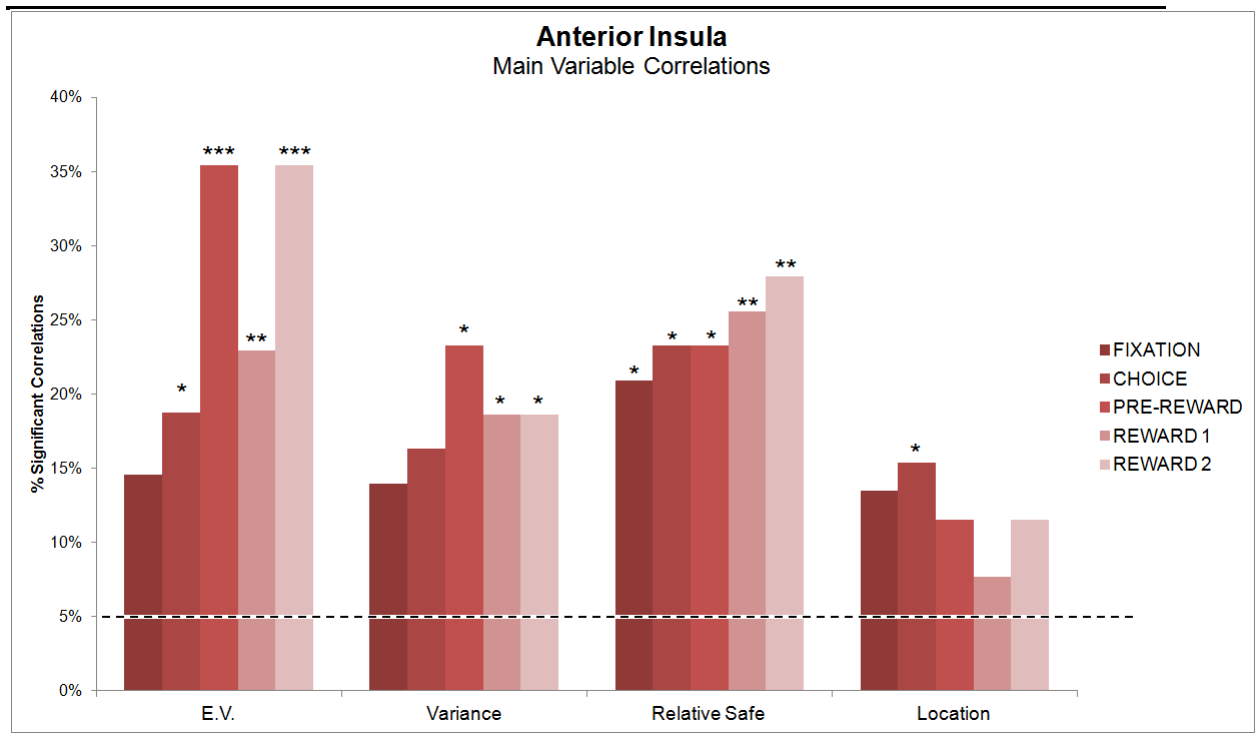


Figure 11. The percentage of significant correlations with the main variables of expected value (E.V.), variance (our measure of risk), relative value of the safe option, and location of saccade choice across the population of anterior insula neurons. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values of each percentage against chance (5% dashed line, one-tailed): *p<0.05, **p<0.01, ***p<0.001.

To further characterize the findings, the main variables (except location) were grouped by the valence of their correlations (positive or negative, see **Figure 12**) and tested for population significance against chance ($p < 0.05$, one-tailed, t-test of percentage against chance). For expected value, there was a significant positive expected value population in the pre-reward and both reward windows (pre-reward: $t(47)=3.07$, $p=0.0018$; reward 1: $t(47)=1.85$, $p=0.035$; reward 2: $t(47)=2.12$, $p=0.02$), and a significant negative expected value population in the pre-reward and reward 2 windows (pre-reward: $t(47)=1.85$, $p=0.035$; reward 2: $t(47)=2.84$, $p=0.0033$). For variance, there was only a significant positive correlation in the reward 1 period ($t(42)=1.93$, $p=0.03$). There was also a significant negative correlation with relative value of the safe option in the choice, pre-reward, and both reward windows (choice: $t(42)=2.7$, $p=0.0045$; pre-reward: $t(42)=1.93$, $p=0.0302$; reward 1 & 2: $t(42)=2.45$, $p=0.0092$).

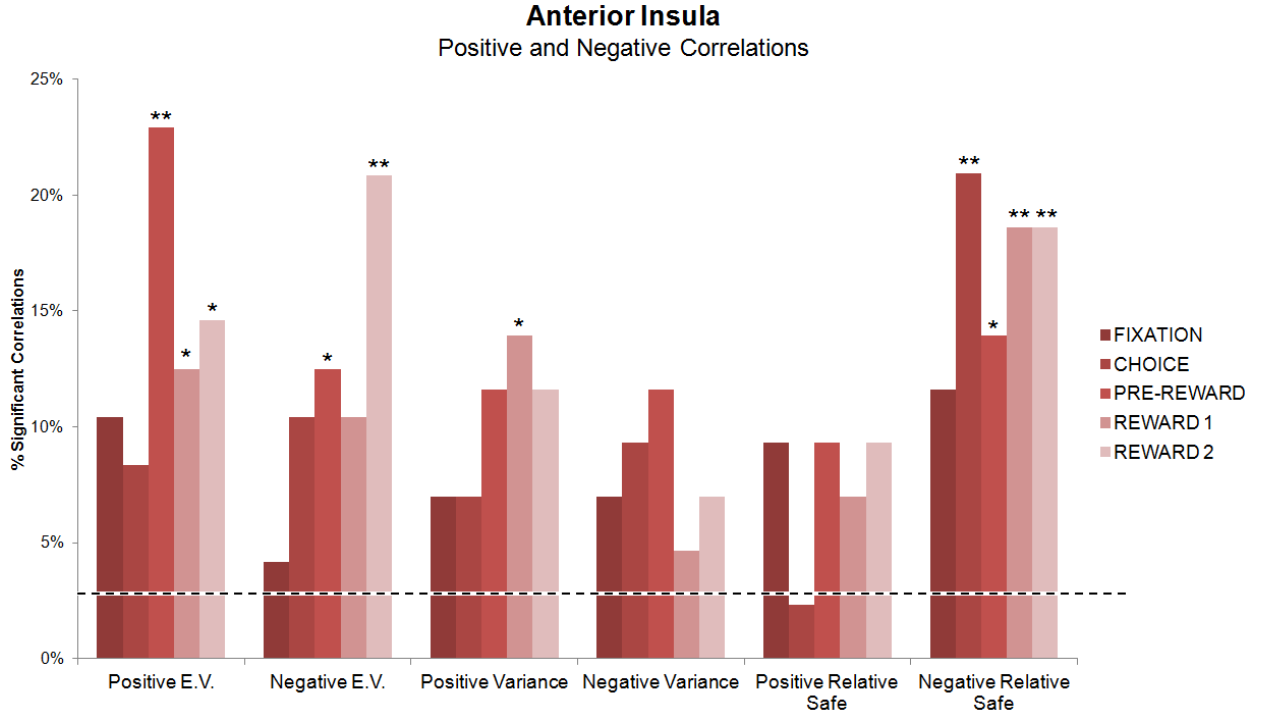


Figure 12. Among the significant correlations, the breakdown percentages of those with positive and negative correlations with the main variables of expected value (E.V.), variance (our measure of risk), and relative value of the safe option across the population of anterior insula neurons. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values of each percentage against chance (2.5% dashed line, one-tailed): * $p < 0.05$, ** $p < 0.01$.

2.4 DISCUSSION

The neurophysiology findings show that throughout the trial, significant populations of anterior insula neurons were correlated with expected value, risk (as measured by variance), the relative value of the safe option, and the location of the options while the monkeys performed the fluid-based gambling task.

For expected value, the correlations did not become significant until after the fixation window, with the highest level of correlations occurring during the pre-reward and reward 2 windows. For risk (variance), significant correlations were found in the pre-reward and reward windows, while relative safe value correlations appeared to be found throughout the trial in all 5 windows. Assessing the correlations by valence, there was a significant positive correlation with expected value during the pre-reward and reward periods and a significant negative correlation with expected value found during both the pre-reward and reward 2 periods. For risk (variance), there was a significant population of positively correlated neurons in the first reward window. And finally there was a significant negative correlation with safe relative value during the choice, pre-reward, and both reward windows. Significant negative correlations were not found for risk and significant positive correlations were not found for relative safe value.

The results found in this study suggest that the insula may process and integrate the positive as well as the negative expected value of rewards and is also correlated with the riskiness of the option (as measured by variance). With respect to valuation and decision-making that may occur in the fixation period, we found no clear evidence for a role of insula cortex. However, there was some evidence for a role in expected value

assessment and action-selection (i.e., from the significant location result) when the actual choice was made. However, the greater role for insula appears to occur after the choice is made, with significant populations of neurons related to expected value and risk in the pre-reward and reward periods. Positive correlations with expected value in these periods suggest involvement in reward anticipation and evaluation. Interestingly, negative correlations with expected value might suggest that a population of neurons in insula produce a type of warning (especially in pre-reward) or error (in reward 2) signal. Note, again, that in reward 2 the risky outcome is resolved.

Our relative safe value variable allowed us to study whether the condition under which the monkey made safe decisions (i.e., the “context”) influenced how the safe option (same stimulus for all conditions) was processed. Significant correlations of relative safe value neurons were found throughout the trial, with only negative correlations showing significance, and only after the fixation period. A negative correlation with the relative value of the safe option indicates that these neurons fired the *least* when the monkey chose the safe option in condition 2 (the safe advantageous condition) and the *most* when the monkey chose the safe option in condition 1 (the safe disadvantageous condition). Along with the significant population of negative expected value neurons, this activity may represent an aversion for the safe option when the risky condition had a higher expected value. Another possibility is that these neurons are influenced by an average overall reward amount in each condition (greatest in condition 1 and lowest in condition 2). This latter possibility would again provide evidence for the relationship of insula neurons to reward in a positive manner.

According to Bechara’s addiction model (2009), the insula holds representations

of pleasurable cues from the environment, such as the place you once smoked. In the fluid-gambling task, this could be the condition 1 risky option stimulus that holds a highly desired 7 squirts of juice. Our significant positive correlations with expected value during pre-reward and reward windows support the addiction model and show that anterior insula neurons may be highly tuned to saliently positive and potentially addictive cues in the environment. Our significant negative correlations for expected value also support the addiction model in the sense that the anterior insula can represent negativity in the environment through urges and cravings for large positive rewards (Garavan, 2010; Naqvi & Bechara, 2009). In the face of less pleasurable rewards (e.g. those with lower expected value, such as the risky option in condition 2), this withdrawal and/or craving signal can become stronger.

The significant populations of insula neurons that correlate with the expected value of options (as opposed to an ‘all or none’ signal of reward, see (Mizuhiki, Richmond, & Shidara, 2012)) supports lesions studies that find insula damage reduces the individual’s ability to integrate value and likelihood and use expected value in decision-making (Clark, et al., 2008; Weller, et al., 2009). Furthermore, our results show that the neurons can process the variance of reward outcomes, consistent with literature implicating its role in risk and uncertainty processing (Paulus, Rogalsky, Simmons, Feinstein, & Stein, 2003; Preusschoff, et al., 2008; Singer, et al., 2009).

Chapter 3.

Neural Processing of Risky Decision-Making by the Premotor Cortex

3.1 PREMOTOR CORTEX

To make decisions, the premotor cortex is central in turning our desires into voluntary actions. Levy and Glimcher (Levy & Glimcher, 2012) recently proposed a schema for decision-making in the brain whereby motor control circuitry such as the premotor cortex integrates sensory and value information in the final representation of choice. Neurons in the premotor cortex have been found to code purposeful motor actions as well as attention toward spatial locations (Bruce & Goldberg, 1985; Coe, Tomihara, Matsuzawa, & Hikosaka, 2002; Schall & Hanes, 1993; Weinrich & Wise, 1982). However, relatively little is known about whether neuronal activity in this brain region correlates with risky decision-making variables such as expected value, risk, and relative value. By recording single neurons from the premotor cortex in the fluid-gambling task, we aim to shed light on this matter.

3.2 NEUROPHYSIOLOGY METHODS

The methods were the same as **2.2 NEUROPHYSIOLOGY METHODS** except neural recordings were made in the premotor cortex (specifically Brodmann areas 6 and 8) of monkeys C and H. Overall, we sampled electrophysiological activity from 48 single premotor cortex neurons (35 & 13 in monkey C & H, respectively) while the monkeys performed the fluid-gambling task (see **Chapter 1bii**). Of these 48 neurons, 31 were stable across all 3 conditions. All 48 neurons were analyzed when appropriate.

3.3 RESULTS

The results for the main variable correlation analysis ($p < 0.05$, one-tailed, t-test of percentage against chance (5%)) are shown in **Figure 13**. Overall, premotor cortex neurons were significantly correlated with all variables in the task in at least one analysis window. For location selectivity, all windows were significantly correlated (fixation and pre-reward: $t(47)=3.53$, $p=0.0005$; choice: $t(47)=3.36$, $p=0.0008$; reward 1: $t(47)=3.15$, $p=0.0015$; reward 2: $t(47)=1.798$, $p=0.039$). Significant correlations with expected value were found in fixation and both reward windows (fixation: $t(44)=2.80$, $p=0.0038$; reward 1: $t(44)=2.11$, $p=0.02$; reward 2: $t(44)=2.35$, $p=0.012$). Variance was significantly correlated in the 2nd reward period only ($t(40)=2.917$, $p=0.0029$). Lastly, there was a significant correlation of relative value of the safe option in all windows except choice (fixation, pre-reward, and reward 1: $t(40)=2.92$, $p=0.0029$; reward 2: $t(40)=2.45$, $p=0.01$).

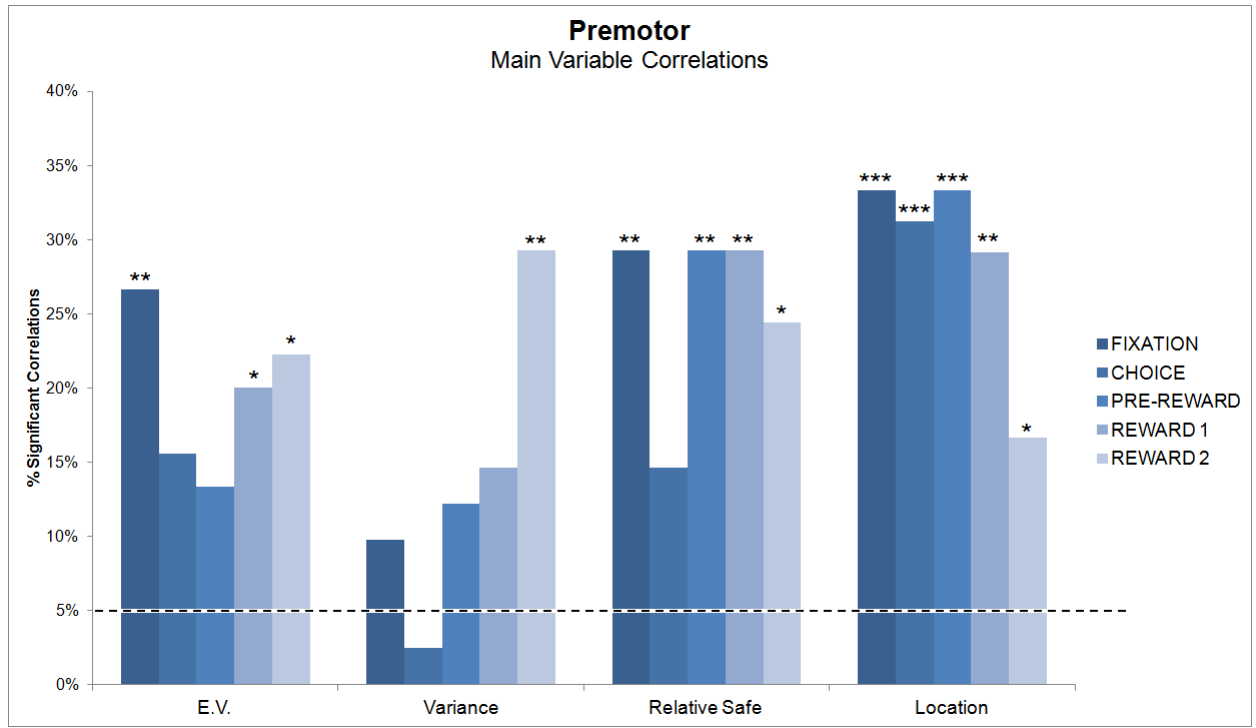


Figure 13. The percentage of significant correlations with the main variables of expected value (E.V.), variance (our measure of risk), relative value of the safe option, and location of saccade choice across the population of premotor cortex neurons. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values of each percentage against chance (5% dashed line, one-tailed): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

To further characterize the findings, the main variables (except location) were grouped by the valence of their correlations (positive or negative, see **Figure 14**) and tested for population significance against chance ($p < 0.05$, one-tailed, t-test of percentage against chance (2.5%)). For correlations with expected value, a significant percentage of positive correlations were found in the fixation window only ($t(44)=1.89$ $p=0.032$), while negative correlations were found in the fixation and reward windows (fixation: $t(44)=1.89$, $p=0.032$; reward 1: $t(44)=2.17$, $p=0.018$; reward 2: $t(44)=2.42$, $p=0.0098$). For variance (our measure of risk), the population of negative correlations in the reward 2 window was significant ($t(40)=2.23$, $p=0.016$). For correlations with the relative value of the safe option, only significant negative correlations were found, and in all analysis windows except choice (fixation & pre-reward: $t(40)=2.23$, $p=0.016$; reward 1: $t(40)=2.75$, $p=0.0045$; reward 2: $t(40)=2.01$, $p=0.026$).

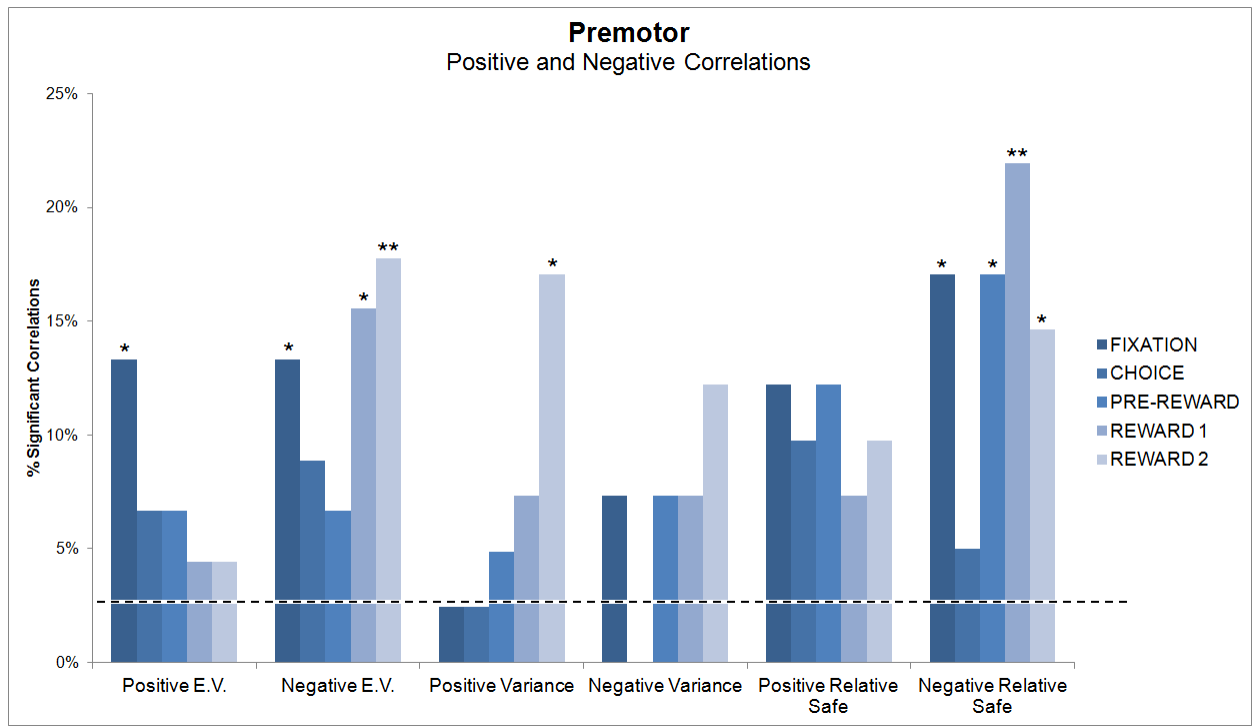


Figure 14. Among the significant correlations, the breakdown percentages of those with positive and negative correlations with the main variables of expected value (E.V.), variance (our measure of risk), and relative value of the safe option across the population of premotor cortex neurons. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values of each percentage against chance (2.5% dashed line, one-tailed): * $p < 0.05$, ** $p < 0.01$.

3.4 DISCUSSION & CONCLUSION

The neural findings show that significant populations of premotor cortex neurons were correlated with expected value, risk (as measured by variance), the relative value of the safe option, and target location while the monkeys performed the fluid-based gambling task.

For expected value, correlations were significant during fixation as well as the two reward windows. The population of risk (as measured by variance) correlations was only significant during the 2nd reward window, while relative safe value correlations appear to be found throughout the trial except in the choice window. Breaking down the correlations by valence, there was a significant population of neurons exhibiting a positive relationship to expected value during fixation, while all other significant valences were negative: for expected value during fixation, and the two reward windows; for variance during reward 2; and for relative safe value in all time periods except choice.

Overall, the results show that the premotor cortex codes expected value of reward as well as its location during fixation, suggesting that this region may participate in valuation, decision-making and action selection. There does not appear to be significant coding of variance, and thus the premotor cortex may process only expected value and not risk during decision-making. During choice, the premotor neurons only appear to reflect location, and thus, they are likely involved in making the choice response. Because the expected value signal was no longer significant in the choice and pre-reward periods, it may no longer be needed for response execution. The expected value signal may also be passed onto areas such as the anterior insula for reward anticipation (in the pre-reward period), and received again from such areas during the reward periods (given

that premotor neurons again show a significant expected value relationship during the two reward periods). The expected value correlations during both reward windows possibly reflect either reward evaluation or a motor preparatory signal so that the monkeys can prepare for the impending number of squirts of juice. However, because the correlations with expected value are significantly negative during the two reward windows, it suggests that these premotor neurons may be preparing for and responding to the worse outcomes.

With respect to risk, there was a significant positive correlation with risk (measured as variance) during the 2nd reward period. Given that the risky option outcome is resolved during this time period, and that activity is highest with the highest variance, it may suggest that premotor neurons are involved in uncertainty evaluation and resolution.

Our relative safe variable allows us to study whether the context or condition in which the safe option is presented influences how it is processed during choice. Significant correlations throughout all windows except choice suggest that neurons in the premotor cortex may code the relative expected value of the safe option depending on what risky option it is compared with. A relative safe signal could reflect “contextual memory” and assist the monkey in choosing advantageously in the three well-learned conditions. In the breakdown analysis of the valence of the correlations, a significant negative correlation was found in all analysis windows except the choice window. A negative relative safe value correlation means that the neurons fired more robustly in condition 1 relative to condition 3 and 2 when choosing from the safe option. This may once again be a sign of ‘dislike’ for the safe option when an advantageous risky option

was present. Alternatively, this activity might reflect a general effect of the overall value or reward rate received in the condition (i.e., based on both the safe and risky options in the condition).

3.4 COMPARISON OF BRAIN AREAS

By comparing the brain regions (anterior insula and premotor cortex), we can gain further insight into their respective functions during affective decision-making. Overall, the population of premotor cortex neurons significantly correlated with expected value and location during fixation, suggesting a possible role in valuation, decision-making and action selection. In contrast, insula cortex showed significant population activity related to expected value and location during the choice period, suggesting that it may also play some role in action-selection. Neither region exhibited a significant population of neurons related to risk assessment during decision-making and choice (i.e., fixation and choice windows), and thus risk assessment during decision-making appears to occur elsewhere (such as in prefrontal cortex). Anterior insula cortex then showed an especially strong relationship to expected value and risk during pre-reward, suggesting a prominent role in outcome anticipation. Both regions then appeared to be involved in outcome evaluation during the reward periods, especially during reward 2, in which the risky outcome is resolved. Interestingly, both regions exhibited a strong negative relationship with expected value during reward 2, suggesting that they both reacted strongly to the more negative outcomes (although the insula also had a significant population of neurons also showing a positive relationship with value as well). Premotor neurons also exhibited a strong positive relationship to variance during reward 2, suggesting a potential role in uncertainty evaluation.

We further made direct comparisons between the regions. As shown in **Figure 15**, the premotor was significantly more location selective compared to the anterior insula

cortex in the fixation, pre-reward, and first reward windows (fixation: $t(98)=2.151$, $p=0.034$; pre-reward: $t(98)=2.625$, $p=0.01$; reward 1: $t(98)=2.80$, $p=0.0062$), as might be expected given the well-established role of premotor cortex in action-selection and spatial attention. Conversely, the anterior insula cortex was significantly more correlated with variance at the choice window ($t(82)=2.168$, $p=0.033$), suggesting a role in risk assessment during choice, as well as with expected value at the pre-reward window ($t(91)=2.47$, $p=0.015$), suggesting a stronger role in reward anticipation. Breaking down the correlations into negative and positive ones (**Figure 16**), we found that the anterior insula had a significantly greater negative correlation for the relative value of the safe option at choice ($t(82)=2.166$, $p=0.033$) as well as a significantly greater positive correlation with expected value at pre-reward ($t(91)=2.144$, $p=0.035$).

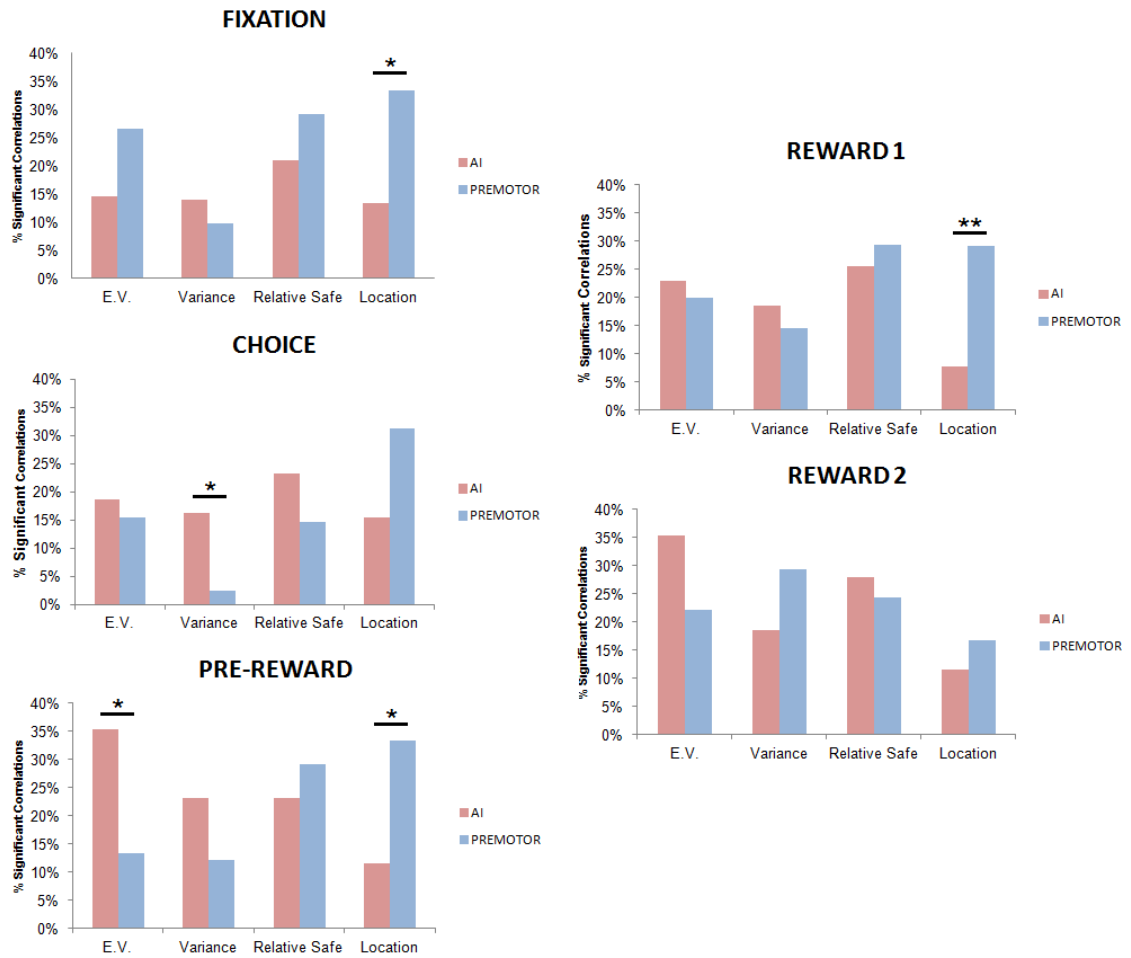


Figure 15. Comparison of the percentage of anterior insula (AI) and premotor neurons showing significant correlations with expected value (E.V.), variance (our measure of risk), the relative value of the safe option, and location. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values using a 2-sample t-test comparing the percentages of significantly correlated neurons: *p<0.05, **p<0.01.

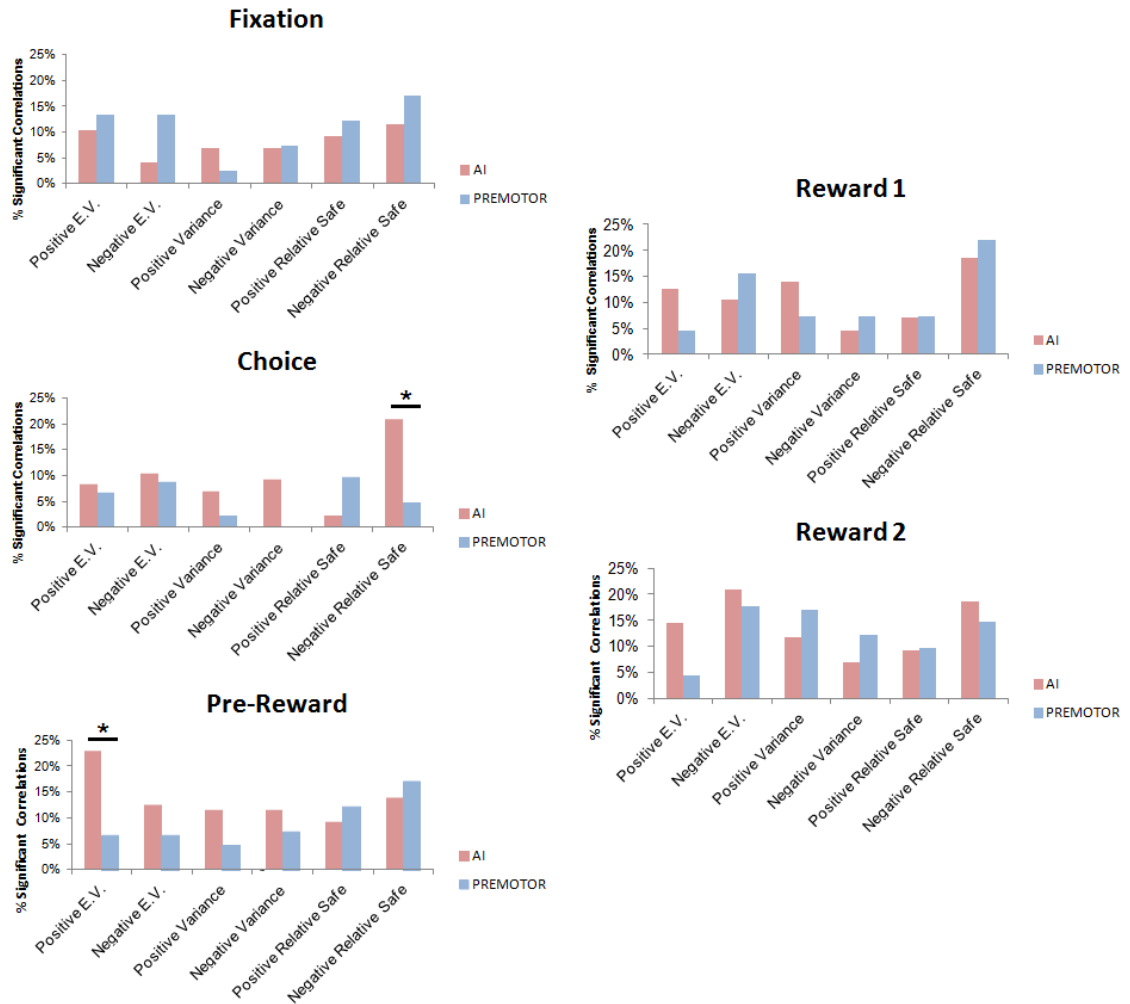


Figure 16. Comparison of the percentage of anterior insula (AI) and premotor neurons showing significant positive or negative correlations with expected value (E.V.), variance (our measure of risk), the relative value of the safe option. Results are shown for each analysis window (fixation, choice, pre-reward, reward 1, and reward 2). Asterisks represent the significant p-values using a 2-sample t-test comparing the percentages of significantly correlated neurons: * $p < 0.05$.

Because the premotor cortex has been shown to code for spatial attention and voluntary action, the results for increased location selectivity in the premotor cortex as compared to anterior insula were expected. It further appears that this location signal remains strong throughout the trial. This persistent coding may help the monkeys remember the locations of what they recently chose and improve learning and performance of the task.

The anterior insula cortex, on the other hand, showed a higher percentage of neurons correlated with variance (a measurement of risk) during the movement period to the target (choice window) as compared to premotor cortex. This confirms previous studies showing this area's significant involvement in decision-making under risk. Moreover, the anterior insula showed a significant positively biased correlation of expected value during the reward anticipation period as compared to premotor cortex. Thus, the anterior insula cortex neurons appear to scale with the mean reward outcome during pre-reward, potentially preparing the individual for the outcome as well as sensitizing learning about outcomes. During the choice analysis window, the anterior insula also shows a larger negative correlation with the relative value of the safe option as compared to the premotor cortex. This suggests that the neurons in the anterior insula fire more while choosing the safe option when it was disadvantageous to do so. This may reflect response conflict or a craving for the large outcome of the risky option.

Altogether, the premotor cortex appears to be more involved in the early stages of the decision-making process, as well as in action control, while the anterior insula encodes risk and location during choice and expected value after the choice is made during reward anticipation. At the same, time neurons in the insula also appear to

produce an opposing signal when the lower outcome is actually obtained (during reward 2). The premotor and anterior insula cortices may work in concert to ensure that reward outcomes and the uncertainty of choice outcomes are evaluated, potentially intensified, and ultimately remembered for future action-selection during decision-making under risk.

REFERENCES

- Admon, R., Bleich-Cohen, M., Weizmant, R., Poyurovsky, M., Faragian, S., & Hendler, T. (2012). Functional and structural neural indices of risk aversion in obsessive-compulsive disorder (OCD). *Psychiatry Res*, 203(2-3), 207-213.
- Ainslie, G. (1975). Specious reward: a behavioral theory of impulsiveness and impulse control. *Psychological Bulletin*, 82(4), 463.
- Baumgartner, H., Sujan, M., & Padgett, D. (1997). Patterns of affective reactions to advertisements: The integration of moment-to-moment responses into overall judgments. *Journal of Marketing Research*, 34(2), 219-232.
- Bechara, A. (2005). Decision making, impulse control and loss of willpower to resist drugs: a neurocognitive perspective. *Nature Neuroscience*, 8(11), 1458-1463.
- Bechara, A., & Damasio, A. R. (2005). The somatic marker hypothesis: A neural theory of economic decision. *Games and Economic Behavior*, 52(2), 336-372.
- Beran, M. J., Ratliff, C. L., & Evans, T. A. (2009). Natural Choice in Chimpanzees (Pan troglodytes): Perceptual and Temporal Effects on Selective Value. *Learn Motiv*, 40(2), 186-196.
- Berntson, G. G., Norman, G. J., Bechara, A., Bruss, J., Tranel, D., & Cacioppo, J. T. (2011). The insula and evaluative processes. *Psychological Science*, 22(1), 80-86.
- Blanchard, T. C., Wolfe, L. S., Vlaev, I., Winston, J. S., & Hayden, B. Y. (2014). Biases in preferences for sequences of outcomes in monkeys. *Cognition*, 130(3), 289-299.
- Bruce, C. J., & Goldberg, M. E. (1985). Primate frontal eye fields. I. Single neurons discharging before saccades. *J Neurophysiol*, 53(3), 603-635.

- Calder, A. J., Keane, J., Manes, F., Antoun, N., & Young, A. W. (2000). Impaired recognition and experience of disgust following brain injury. *Nature Neuroscience*, 3(11), 1077-1078.
- Caraco, T., Martindale, S., & Whittam, T. S. (1980). An empirical demonstration of risk-sensitive foraging preferences. *Animal Behaviour*, 28(3), 820-830.
- Caruana, F., Jezzini, A., Sbriscia-Fioretti, B., Rizzolatti, G., & Gallese, V. (2011). Emotional and Social Behaviors Elicited by Electrical Stimulation of the Insula in the Macaque Monkey. *Current Biology*, 21(3), 195-199.
- Castro, C. A., & Larsen, T. (1992). Primacy and recency effects in nonhuman primates. *Journal of Experimental Psychology: Animal Behavior Processes*, 18(4), 335.
- Chen, M. K., Lakshminarayanan, V., & Santos, L. R. (2006). How basic are behavioral biases? Evidence from capuchin monkey trading behavior. *Journal of Political Economy*, 114(3), 517-537.
- Chikama, M., McFarland, N. R., Amaral, D. G., & Haber, S. N. (1997). Insular cortical projections to functional regions of the striatum correlate with cortical cytoarchitectonic organization in the primate. *Journal of Neuroscience*, 17(24), 9686-9705.
- Clark, L., Bechara, A., Damasio, H., Aitken, M. R. F., Sahakian, B. J., & Robbins, T. W. (2008). Differential effects of insular and ventromedial prefrontal cortex lesions on risky decision-making. *Brain*, 131, 1311-1322.
- Coe, B., Tomihara, K., Matsuzawa, M., & Hikosaka, O. (2002). Visual and anticipatory bias in three cortical eye fields of the monkey during an adaptive decision-making task. *The Journal of neuroscience*, 22(12), 5081-5090.
- Contreras, M., Ceric, F., & Torrealba, F. (2007). Inactivation of the interoceptive insula disrupts drug craving and malaise induced by lithium. *Science*, 318(5850), 655-658.
- Craig, A. (2009). How do you feel—now? the anterior insula and human awareness.
- Critchley, H. D. (2009). Psychophysiology of neural, cognitive and affective integration: fMRI and autonomic indicants. *International Journal of Psychophysiology*, 73(2), 88-94.

- Damasio, A. R., Grabowski, T. J., Bechara, A., Damasio, H., Ponto, L. L., Parvizi, J., et al. (2000). Subcortical and cortical brain activity during the feeling of self-generated emotions. *Nat Neurosci*, 3(10), 1049-1056.
- Davis, C., Levitan, R. D., Muglia, P., Bewell, C., & Kennedy, J. L. (2004). Decision-making deficits and overeating: a risk model for obesity. *Obes Res*, 12(6), 929-935.
- Delgado, M. R., Gillis, M. M., & Phelps, E. A. (2008). Regulating the expectation of reward via cognitive strategies. *Nat Neurosci*, 11(8), 880-881.
- Do, A. M., Rupert, A. V., & Wolford, G. (2008). Evaluations of pleasurable experiences: The peak-and-end rule. *Psychonomic Bulletin & Review*, 15(1), 96-98.
- Dolan, R. J., & Dayan, P. (2013). Goals and Habits in the Brain. *Neuron*, 80(2), 312-325.
- de Abreu, F. B., & Kacelnik, A. (1999). Energy budgets and risk-sensitive foraging in starlings. *Behavioral Ecology*, 10(3), 338-345.
- Evans, J. S. B., & Stanovich, K. E. (2013). Dual-process theories of higher cognition advancing the debate. *Perspectives on Psychological Science*, 8(3), 223-241.
- Evans, T. A., & Beran, M. J. (2007). Delay of gratification and delay maintenance by rhesus macaques (*Macaca mulatta*). *Journal of General Psychology*, 134(2), 199-216.
- Fiorillo, C. D., Tobler, P. N., & Schultz, W. (2003). Discrete coding of reward probability and uncertainty by dopamine neurons. *Science*, 299(5614), 1898-1902.
- Fishbein, D. H., Eldreth, D. L., Hyde, C., Matochik, J. A., London, E. D., Contoreggi, C., et al. (2005). Risky decision making and the anterior cingulate cortex in abstinent drug abusers and nonusers. *Brain Res Cogn Brain Res*, 23(1), 119-136.
- Flaherty, C. F. (1999). *Incentive relativity* (Vol. 15): Cambridge University Press.
- Flaherty, C. F., Turovsky, J., & Krauss, K. L. (1994). Relative hedonic value modulates anticipatory contrast. *Physiology & Behavior*, 55(6), 1047-1054.
- Fredrickson, B. L., & Kahneman, D. (1993). Duration neglect in retrospective evaluations of affective episodes. *Journal of personality and social psychology*, 65(1), 45.

- Fudge, J. L., Breitbart, M. A., Danish, M., & Pannoni, V. (2005). Insular and gustatory inputs to the caudal ventral striatum in primates. *Journal of Comparative Neurology*, 490(2), 101-118.
- Garavan, H. (2010). Insula and drug cravings. *Brain structure and function*, 214(5-6), 593-601.
- Gaspar, P., Bloch, B., & Le Moine, C. (1995). D1 and D2 receptor gene expression in the rat frontal cortex: cellular localization in different classes of efferent neurons. *Eur J Neurosci*, 7(5), 1050-1063.
- Genovesio, A., Brasted, P. J., Mitz, A. R., & Wise, S. P. (2005). Prefrontal cortex activity related to abstract response strategies. *Neuron*, 47(2), 307-320.
- Hare, T. A., Camerer, C. F., & Rangel, A. (2009). Self-control in decision-making involves modulation of the vmPFC valuation system. *Science*, 324(5927), 646-648.
- Hassin, R. R., Ochsner, K. N., & Trope, Y. (2010). *Self control in society, mind, and brain*. Oxford ; New York: Oxford University Press.
- Hayden, B. Y., Heilbronner, S. R., Nair, A. C., & Platt, M. L. (2008). Cognitive influences on risk-seeking by rhesus macaques. *Judgment and Decision Making*, 3(5), 389-395.
- Hayden, B. Y., & Platt, M. L. (2007). Temporal discounting predicts risk sensitivity in rhesus Macaques. *Current Biology*, 17(1), 49-53.
- Hayden, B. Y., & Platt, M. L. (2009). Gambling for Gatorade: risk-sensitive decision making for fluid rewards in humans. *Animal Cognition*, 12(1), 201-207.
- Heilbronner, S. R., & Hayden, B. Y. (2013). Contextual factors explain risk-seeking preferences in rhesus monkeys. *Front Neurosci*, 7, 7.
- Heilbronner, S. R., Rosati, A. G., Stevens, J. R., Hare, B., & Hauser, M. D. (2008). A fruit in the hand or two in the bush? Divergent risk preferences in chimpanzees and bonobos. *Biology Letters*, 4(3), 246-249.

- Heino, A., vanderMolen, H. H., & Wilde, G. J. S. (1996). Differences in risk experience between sensation avoiders and sensation seekers. *Personality and Individual Differences*, 20(1), 71-79.
- Hertwig, R., Barron, G., Weber, E. U., & Erev, I. (2004). Decisions from experience and the effect of rare events in risky choice. *Psychological Science*, 15(8), 534-539.
- Hsee, C. K. (1998). Less is better: When low-value options are valued more highly than high-value options. *Journal of Behavioral Decision Making*, 11(2), 107-121.
- Hsee, C. K., Loewenstein, G. F., Blount, S., & Bazerman, M. H. (1999). Preference reversals between joint and separate evaluations of options: A review and theoretical analysis. *Psychological Bulletin*, 125(5), 576.
- Hsee, C. K., & Zhang, J. (2004). Distinction bias: misprediction and mischoice due to joint evaluation. *Journal of personality and social psychology*, 86(5), 680.
- Ishii, H., Ohara, S., Tobler, P. N., Tsutsui, K., & Iijima, T. (2012). Inactivating anterior insular cortex reduces risk taking. *Journal of Neuroscience*, 32(45), 16031-16039.
- Jeong, J., Oh, Y., Chun, M., & Kralik, J. D. (2014). Preference-Based Serial Decision Dynamics: Your First Sushi Reveals Your Eating Order at the Sushi Table. *Plos One*, 9(5), e96653.
- Jung, K., & Kralik, J. D. (2013). Get It While It's Hot: A Peak-First Bias in Self-Generated Choice Order in Rhesus Macaques. *Plos One*, 8(12), e83814.
- Kacelnik, A., & Bateson, M. (1996). Risky theories - The effects of variance on foraging decisions. *American Zoologist*, 36(4), 402-434.
- Kacelnik, A., & Bateson, M. (1997). Risk-sensitivity: crossroads for theories of decision-making. *Trends in Cognitive Sciences*, 1(8), 304-309.
- Kahneman, D. (2011). *Thinking, fast and slow*. New York: Farrar, Straus, and Giroux.
- Kahneman, D., & Frederick, S. (2002). Representativeness revisited: Attribute substitution in intuitive judgment. In T. Gilovich, D. W. Griffin & D. Kahneman (Eds.), *Heuristics and*

- biases: The psychology of intuitive judgment* (pp. 49-81). Cambridge, England: Cambridge University Press.
- Kahneman, D., Fredrickson, B. L., Schreiber, C. A., & Redelmeier, D. A. (1993). When more pain is preferred to less: Adding a better end. *Psychological Science*, 4(6), 401-405.
- Kahneman, D., Slovic, P., & Tversky, A. (Eds.). (1982). *Judgement under uncertainty: Heuristics and biases*. Cambridge: Cambridge University Press.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: Analysis of decision under risk. *Econometrica*, 47(2), 263-291.
- Kahneman, D., & Tversky, A. (1984). Choices, values, and frames. *American Psychologist*, 39(4), 341-350.
- Kahneman, D., & Tversky, A. (2000). *Choices, values, and frames*. New York: Cambridge, UK: Russell sage Foundation ; Cambridge University Press.
- Klos, A. (2005). Investment decisions and time horizon: Risk perception and risk behavior in repeated gambles. *Management Science*, 51(12), 1777-1790.
- Knutson, B., Rick, S., Wimmer, G. E., Prelec, D., & Loewenstein, G. (2007). Neural predictors of purchases. *Neuron*, 53(1), 147-156.
- Kober, H., Mende-Siedlecki, P., Kross, E. F., Weber, J., Mischel, W., Hart, C. L., et al. (2010). Prefrontal-striatal pathway underlies cognitive regulation of craving. *Proc Natl Acad Sci U S A*, 107(33), 14811-14816.
- Kralik, J. D., & Sampson, W. W. (2012). A fruit in hand is worth many more in the bush: steep spatial discounting by free-ranging rhesus macaques (*Macaca mulatta*). *Behav Processes*, 89(3), 197-202.
- Kralik, J. D., Xu, E. R., Knight, E. J., Khan, S. A., & Levine, W. J. (2012). When Less Is More: Evolutionary Origins of the Affect Heuristic. *Plos One*, 7(10), e46240.

- Kreek, M. J., Nielsen, D. A., Butelman, E. R., & LaForge, K. S. (2005). Genetic influences on impulsivity, risk taking, stress responsivity and vulnerability to drug abuse and addiction. *Nat Neurosci*, 8(11), 1450-1457.
- Kuhnen, C. M., & Knutson, B. (2005). The neural basis of financial risk taking. *Neuron*, 47(5), 763-770.
- Lakshminarayanan, V. R., Chen, M. K., & Santos, L. R. (2011). The evolution of decision-making under risk: Framing effects in monkey risk preferences. *Journal of Experimental Social Psychology*, 47(3), 689-693.
- Levy, D. J., & Glimcher, P. W. (2012). The root of all value: a neural common currency for choice. *Current Opinion in Neurobiology*, 22(6), 1027-1038.
- List, J. A. (2002). Preference reversals of a different kind: The "More is less" Phenomenon. *American Economic Review*, 1636-1643.
- Logue, A. W. (1995). *Self-control: Waiting until tomorrow for what you want today*: Prentice-Hall, Inc.
- Mantonakis, A., Rodero, P., Lesschaeve, I., & Hastie, R. (2009). Order in Choice Effects of Serial Position on Preferences. *Psychological Science*, 20(11), 1309-1312.
- McClure, S. M., Laibson, D. I., Loewenstein, G., & Cohen, J. D. (2004). Separate neural systems value immediate and delayed monetary rewards. *Science*, 306(5695), 503-507.
- McCoy, A. N., & Platt, M. L. (2005). Risk-sensitive neurons in macaque posterior cingulate cortex. *Nat Neurosci*, 8(9), 1220-1227.
- Mizuhiki, T., Richmond, B. J., & Shidara, M. (2012). Encoding of reward expectation by monkey anterior insular neurons. *J Neurophysiol*, 107(11), 2996-3007.
- Montgomery, H., & Adelbratt, T. (1982). Gambling decisions and information about expected value. *Organizational Behavior and Human Performance*, 29(1), 39-57.
- Naqvi, N. H., & Bechara, A. (2009). The hidden island of addiction: the insula. *Trends in Neurosciences*, 32(1), 56-67.

- Naqvi, N. H., Rudrauf, D., Damasio, H., & Bechara, A. (2007). Damage to the insula disrupts addiction to cigarette smoking. *Science*, 315(5811), 531-534.
- Niv, Y., Daw, N. D., Joel, D., & Dayan, P. (2007). Tonic dopamine: opportunity costs and the control of response vigor. *Psychopharmacology*, 191(3), 507-520.
- O'Neill, M., & Schultz, W. (2010). Coding of reward risk by orbitofrontal neurons is mostly distinct from coding of reward value. *Neuron*, 68(4), 789-800.
- Passingham, R. E., & Wise, S. P. (2012). *The neurobiology of the prefrontal cortex: Anatomy, evolution, and the origin of insight*. Oxford UK: Oxford University Press.
- Paulsen, D. J., Carter, R. M., Platt, M. L., Huettel, S. A., & Brannon, E. M. (2011). Neurocognitive development of risk aversion from early childhood to adulthood. *Front Hum Neurosci*, 5, 178.
- Paulsen, D. J., Platt, M. L., Huettel, S. A., & Brannon, E. M. (2011). Decision-making under risk in children, adolescents, and young adults. *Front Psychol*, 2, 72.
- Paulsen, D. J., Platt, M. L., Huettel, S. A., & Brannon, E. M. (2012). From risk-seeking to risk-averse: the development of economic risk preference from childhood to adulthood. *Front Psychol*, 3, 313.
- Paulus, M. P., Rogalsky, C., Simmons, A., Feinstein, J. S., & Stein, M. B. (2003). Increased activation in the right insula during risk-taking decision making is related to harm avoidance and neuroticism. *Neuroimage*, 19(4), 1439-1448.
- Paxinos, G., Huang, X. F., & Toga, A. W. (2000). *The rhesus monkey brain in stereotaxic coordinates*. San Diego, CA: Academic Press.
- Preuschoff, K., Quartz, S. R., & Bossaerts, P. (2008). Human insula activation reflects risk prediction errors as well as risk. *Journal of Neuroscience*, 28(11), 2745-2752.
- Rabin, M., & Vayanos, D. (2010). The Gambler's and Hot-Hand Fallacies: Theory and Applications. *Review of Economic Studies*, 77(2), 730-778.

- Rachlin, H., & Green, L. (1972). COMMITMENT, CHOICE AND SELF-CONTROL¹. *Journal of the experimental analysis of behavior*, 17(1), 15-22.
- Rasmussen, S. A., & Eisen, J. L. (1992). The epidemiology and clinical features of obsessive compulsive disorder. *Psychiatr Clin North Am*, 15(4), 743-758.
- Redelmeier, D. A., & Kahneman, D. (1996). Patients' memories of painful medical treatments: Real-time and retrospective evaluations of two minimally invasive procedures. *Pain*, 66(1), 3-8.
- Redelmeier, D. A., Katz, J., & Kahneman, D. (2003). Memories of colonoscopy: a randomized trial. *Pain*, 104(1-2), 187-194.
- Richfield, E. K., Young, A. B., & Penney, J. B. (1989). Comparative distributions of dopamine D-1 and D-2 receptors in the cerebral cortex of rats, cats, and monkeys. *J Comp Neurol*, 286(4), 409-426.
- Rode, E., Rozin, P., & Durlach, P. (2007). Experienced and remembered pleasure for meals: Duration neglect but minimal peak, end (recency) or primacy effects. *Appetite*, 49(1), 18-29.
- Rosati, A. G., Stevens, J. R., Hare, B., & Hauser, M. D. (2007). The evolutionary origins of human patience: temporal preferences in chimpanzees, bonobos, and human adults. *Current Biology*, 17(19), 1663-1668.
- Rothschild, M., & Stiglitz, J. E. (1970). Increasing risk: I. A definition. *Journal of Economic theory*, 2(3), 225-243.
- Samuelson, P. A. (1963). Risk and uncertainty: A fallacy of large numbers. *Scientia*, 98(612), 108-&.
- Schall, J. D., & Hanes, D. P. (1993). Neural basis of saccade target selection in frontal eye field during visual search. *Nature*, 366(6454), 467-469.
- Schreiber, C. A., & Kahneman, D. (2000). Determinants of the remembered utility of aversive sounds. *Journal of Experimental Psychology-General*, 129(1), 27-42.

- Schuck-Paim, C., Pompilio, L., & Kacelnik, A. (2004). State-dependent decisions cause apparent violations of rationality in animal choice. *Plos Biology*, 2(12), 2305-2315.
- Silberberg, A., Widholm, J. J., Bresler, D., Fujita, K., & Anderson, J. R. (1998). Natural choice in nonhuman primates. *J Exp Psychol Anim Behav Process*, 24(2), 215-228.
- Singer, T., Critchley, H. D., & Preuschoff, K. (2009). A common role of insula in feelings, empathy and uncertainty. *Trends Cogn Sci*, 13(8), 334-340.
- Soane, E., & Chmiel, N. (2005). Are risk preferences consistent? The influence of decision domain and personality. *Personality and Individual Differences*, 38(8), 1781-1791.
- Stevens, J. R., Hallinan, E. V., & Hauser, M. D. (2005). The ecology and evolution of patience in two New World monkeys. *Biology Letters*, 1(2), 223-226.
- Sundali, J., & Croson, R. (2006). Biases in casino betting: The hot hand and the gambler's fallacy. *Judgment and Decision Making Journal*, 1(1), 1-12.
- Thierry, B., Mewa, S., & Kaumanns, W. (2004). *Macaque societies : a model for the study of social organization*. Cambridge ; New York, NY: Cambridge University Press.
- Tobler, P. N., Fiorillo, C. D., & Schultz, W. (2005). Adaptive coding of reward value by dopamine neurons. *Science*, 307(5715), 1642-1645.
- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157), 1124-1131.
- Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211(4481), 453-458.
- Volkow, N. D. (2010). Addiction: Conflict Between Brain Circuits. *Biological Psychiatry*, 67(9), 93s-94s.
- Volkow, N. D., Fowler, J. S., Wang, G. J., Telang, F., Logan, J., Jayne, M., et al. (2010). Cognitive control of drug craving inhibits brain reward regions in cocaine abusers. *Neuroimage*, 49(3), 2536-2543.

- Weber, E. U., Blais, A. R., & Betz, N. E. (2002). A domain-specific risk-attitude scale: Measuring risk perceptions and risk behaviors. *Journal of Behavioral Decision Making*, 15(4), 263-+.
- Weber, E. U., Shafir, S., & Blais, A.-R. (2004). Predicting risk sensitivity in humans and lower animals: risk as variance or coefficient of variation. *Psychological review*, 111(2), 430.
- Weinrich, M., & Wise, S. P. (1982). The premotor cortex of the monkey. *The Journal of neuroscience*, 2(9), 1329-1345.
- Weller, J. A., Levin, I. P., Shiv, B., & Bechara, A. (2009). The effects of insula damage on decision-making for risky gains and losses. *Social Neuroscience*, 4(4), 347-358.
- Weller, J. A., & Tikir, A. (2011). Predicting Domain-Specific Risk Taking With the HEXACO Personality Structure. *Journal of Behavioral Decision Making*, 24(2), 180-201.
- Williams, B. A. (1997). Varieties of contrast: A review of Incentive Relativity by Charles F. Flaherty. *Journal of the experimental analysis of behavior*, 68(1), 133-141.
- Williams, B. A. (2002). Behavioral contrast redux. *Animal Learning & Behavior*, 30(1), 1-20.
- Wilson, T. D., & Gilbert, D. T. (2005). Affective forecasting - Knowing what to want. *Current Directions in Psychological Science*, 14(3), 131-134.
- Wise, S. P. (2008). Forward frontal fields: phylogeny and fundamental function. *Trends in Neurosciences*, 31(12), 599-608.
- Wright, A. A., Santiago, H. C., & Sands, S. F. (1984). Monkey memory: Same/different concept learning, serial probe acquisition, and probe delay effects. *Journal of Experimental Psychology: Animal Behavior Processes*, 10(4), 513.
- Wright, A. A., Santiago, H. C., Sands, S. F., Kendrick, D. F., & Cook, R. G. (1985). Memory processing of serial lists by pigeons, monkeys, and people. *Science*, 229(4710), 287-289.
- Xue, G., Lu, Z. L., Levin, I. P., & Bechara, A. (2010). The impact of prior risk experiences on subsequent risky decision-making: The role of the insula. *Neuroimage*, 50(2), 709-716.

Yacubian, J., Sommer, T., Schroeder, K., Glascher, J., Braus, D. F., & Buchel, C. (2007).

Subregions of the ventral striatum show preferential coding of reward magnitude and probability. *Neuroimage*, 38(3), 557-563.

Yamada, H., Tymula, A., Louie, K., & Glimcher, P. W. (2013). Thirst-dependent risk preferences in monkeys identify a primitive form of wealth. *Proceedings of the National Academy of Sciences*, 110(39), 15788-15793.

Zentall, T. R., & Singer, R. A. (2007). WITHIN-TRIAL CONTRAST: PIGEONS PREFER
CONDITIONED REINFORCERS THAT FOLLOW A RELATIVELY MORE
RATHER THAN A LESS AVERSIVE EVENT. *Journal of the experimental analysis of
behavior*, 88(1), 131-149.